

SIEMENS

MAGNETOM

MR

Replacements of Parts

System

Host / Imager

MAGNETOM Verio

© Siemens 2008

The reproduction, transmission or use of this document or its contents is not permitted without express written authority. Offenders will be liable for damages. All rights, including rights created by patent grant or registration of a utility model or design, are reserved.

Print No.: M6-040.841.12.03.02
Replaces: M6-040.841.12.02.02

English
Doc. Gen. Date: 01.09

Document revision level

The document corresponds to the version/revision level effective at the time of system delivery. Revisions to hardcopy documentation are not automatically distributed.

Please contact your local Siemens office to order current revision levels.

Disclaimer

The installation and service of equipment described herein is to be performed by qualified personnel who are employed by Siemens or one of its affiliates or who are otherwise authorized by Siemens or one of its affiliates to provide such services.

Assemblers and other persons who are not employed by or otherwise directly affiliated with or authorized by Siemens or one of its affiliates are directed to contact one of the local offices of Siemens or one of its affiliates before attempting installation or service procedures.

1	HOST	5
	General	5
	Safety information	5
	FRU overview	6
	DVD-RW drive	7
	Required time and tools	7
	Preliminary	7
	Disassembly	7
	Assembling	8
	Final checks	8
	Host	9
	Required time and tools	9
	Preliminary	9
	Disassembly	9
	Assembling	9
	Final checks	9
	Dongle	10
	Required time and tools	10
	Preliminary	10
	Disassembly	10
	Assembling	10
	Final checks	11
	Fans, ventilation duct, crosspiece (R640 / R650)	12
	Front fan	12
	Side fan	15
	Ventilation duct, CPU fan and rear fan	17
	Crosspiece	21
	Power supply (R640 / R650)	23
	Preparatory work steps	23
	Changing the power supply	23
	Final work steps	25
	Battery	26
	Changing the lithium battery	26
	Final work steps	27
	In-Room MRC	28
	Required time and tools	28
	Final checks	28
2	Imager	29
	General	29
	Safety information	29
	FRU overview	30
	CPU or HD container (High end)	31
	Required time and tools	31
	Preliminary	31
	Disassembly	31

Assembling	32
Final checks	32
Imager (low end)	33
Required time and tools	33
Preliminary	33
Disassembly	33
Assembling	33
Final checks	34
3 TFT color monitors	35
General information	35
Tools and time required	35
Preliminary	36
Disassembly	36
Assembly	36
Final steps	36
Safety	37
Digital TFT monitor DSC 1904-D	38
Connections	38
Adjustments	41
Digital TFT monitor DSC 1907-D	47
Connections	47
Adjustments	50
Digital TFT monitor DSC 1907-D Rev02	57
Connections	57
Adjustments	61
Video cabling for digital monitors	68
With additional options and digital/analog signal transmission	68
Start-up procedure for additional monitors (optional)	73
Clone Mode	74
Appendix	77
Visually inspecting/evaluating the TFT monitor	77
1. Remedy for misadjustment of brightness / contrast	87
2. Resetting the factory defaults	88
3. Missing color, color overtone or incorrect setting for screen color	89
4. Automatic adjustment of geometry failed (for monitor type 'B' only)	89
4 BIOS CD	90
BIOS CD update and installation	90
General	90
Prerequisites	90
BIOS update on MRC	91
BIOS update of the MRIR	92
Final checks	93
5 Changes to previous version	94

General

Safety information



Adhere to ESD regulations and use the appropriate tool set: anti-static wristband and grounded tabletop pad (TDK equipment Type 8005 from 3M Deutschland GmbH).

FRU overview

FRU	Time	Instructions	Final checks	Tools
DVD-RW	45 min.	(DVD-RW drive / p. 7)	Archive a patient images on CD-R / DVD media with the new DVD-RW drive	Field service grounding kit (ESD), set of standard tools
Host	180 min.	(Host / p. 9)	Perform a measurement, view the images produced. Check with the customer that all options are available	Field service grounding kit (ESD), set of standard tools
Dongle (USB)	60 min.	(Dongle / p. 10)	Check with the customer that all options are available Perform a measurement and view the images produced.	Service license/Application licenses. The Service/Applications licenses generated in relation to the flexid of the new dongle
Fans	45 min.	(Fans, ventilation duct, crosspiece (R640 / R650) / p. 12)	Ensure that the new fan(s) works.	Set of standard tools
Power supply	45 min.	(Power supply (R640 / R650) / p. 23)	Perform a measurement and view the images produced.	Set of standard tools
Battery	60 min.	(Battery / p. 26)	Perform BIOS setup according to TDOC Installation Software. Perform a measurement and view the images produced.	Field service grounding kit (ESD), set of standard tools
In-room MRC	120 min.	(In-Room MRC / p. 28) Safety-related tests	Run the auto configuration program and perform a visual evaluation according to the start-up of the In-Room MRC Measure protective conductor resistance for the "In-room MRC"	Set of standard tools and TDOC start-up of the In-room MRC

DVD-RW drive

Required time and tools

Time

- 45 min.

Tools

- Set of standard tools

Preliminary

- Shutdown and switch off the system.

Disassembly

Disconnect the main power plug, network drop cable(s), mouse, and keyboard cables. Remove the right side cover (as seen from back), remove the 2 screws for the side cover. Slide the side cover 15 mm back, and remove the cover laterally.

⚠ Adhere to ESD regulations and use the appropriate tool set: anti-static wristband and grounded tabletop pad (TDK equipment Type 8005 from 3M Deutschland GmbH).



- Remove the plastic front cover to access the CD-RW drive. Gently bend the plastic front cover clips, and pull the cover forward

Fig. 1: Host "Celsius R610", front cover



Fig. 2: Host, disconnect IDE and power cable

- Disconnect the IDE and power cables in back of the CD-RW drive

Press the metallic clips located along the sides at the front of the DVD-RW drive. Gently pull out the DVD-RW drive.

Only valid for IDE devices; the new DVD-RW drive must be jumpered as “Master” for the IDE bus. Refer to the label on the DVD-RW drive. (Since R650 S-ATA DVD-RW drives are used)

Assembling

- Reassemble the “host” with the new DVD-RW drive

Final checks

- Switch on the system, start the MR application
- Press **Ctrl - Esc**; select **Start, Advanced User**; enter the advanced user password; click **OK**
- Check that drive letters for the CD writer and DVD-drive are correct with the “windows file explorer” (CD/DVD-burner drive letter “**S**” and DVD/CD-ROM drive letter “**R**”). (In case of wrong drive letters, change the drive letter with the disk manager as follows: Click **Start; Run; diskmgmt.msc** ↵ ; **right click** the drive to be corrected; select **Change Driveletters and Path...** select proper drive letter in the pull-down menu, confirm the pop-up with **Yes**, repeat this sequence until you active the correct drive letter assignment)
- Archive patient images on CD-R / DVD media with the new DVD-RW drive

Host

Required time and tools

Time

- 180 min.

Tools

- Set of standard tools

Preliminary

- If the condition of the “Host” allows access to the “Service platform” (Service Software, e.g., as user administrator), select **Backup & Restore** in the “Service Software”. Back up the following package:
 - **Master Backup**
- Shutdown and switch off the system.

Disassembly

NOTE

Remember to move the dongle from the old “host” to the new “host”

- Disconnect the main power plug, network drop cable(s), mouse, and keyboard cables.

Assembling

- Plug the HW dongle to the new Host.
- Plug the network drop cable(s), mouse, keyboard cables, and main power plug.

Final checks

- Check if the new “Host” has the correct software version or install software on the new “Host” according to the technical documentation (TDOC) “Install Software” valid for your system.
- Restore “site specific data” according to TDOC “Install Software”, valid for your system.
- Start the Local Service, select **Quality Assurance**, Perform the “Quality Assurance” step Image Orientation

Dongle

Required time and tools

Time

- 60 min.

Tools

- Licence file with all customer licenses for the new DONGLE
- Valid service license for the new DONGLE

Preliminary

If the service license and all customer options are invalid, the dongle is most likely defective.

NOTE

Place dongle orders only through TDHSC1 in Erlangen, a dongle defect must be directly escalated to the TDHSC1 in Erlangen, phone number:+49 9191-188080

Disassembly



Fig. 3: Host, USB dongle

- Ensure that the “Host” is switched off; replace the dongle in back of the “Host”.

Assembling

- Switch on the system

Final checks

- Login as user **administrator**, start the Internet Explorer, start Local Service, select **Configuration**, Service, **Licensing**, Click radio button **Browse to select the license file**, navigate to the media with your new license file, click **Save**.
- Re-boot the system, start the MR application
- Check with the customer that all options are available
- Start the Local service, select **Backup & Restore** in the "Service Software". Back up the following package:
 - **Master Backup**

Fans, ventilation duct, crosspiece (R640 / R650)

Front fan

- Open the housing.
- Open the front panel.

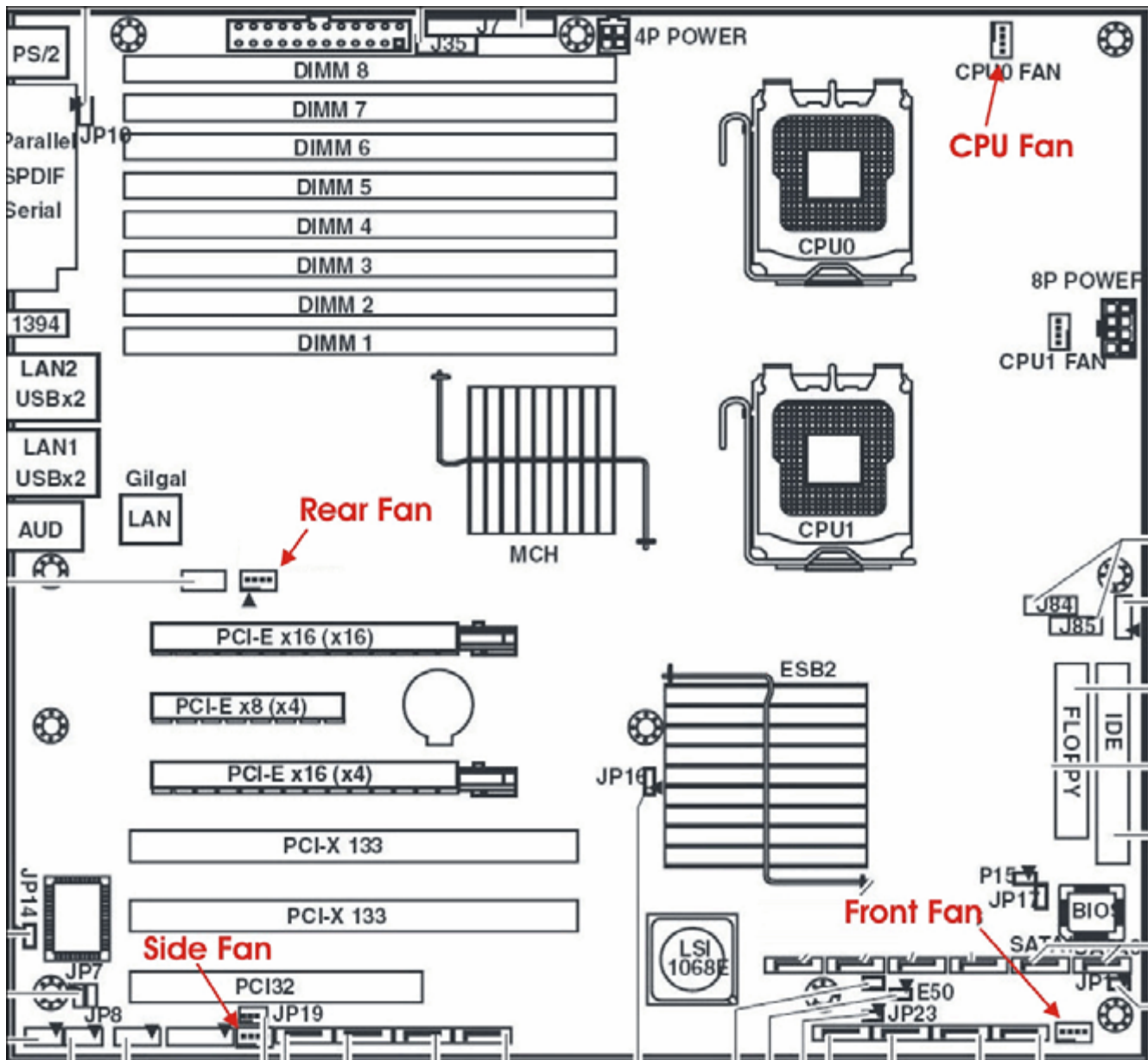


Fig. 4: Fan connectors on the mainboard

Removing the front fan

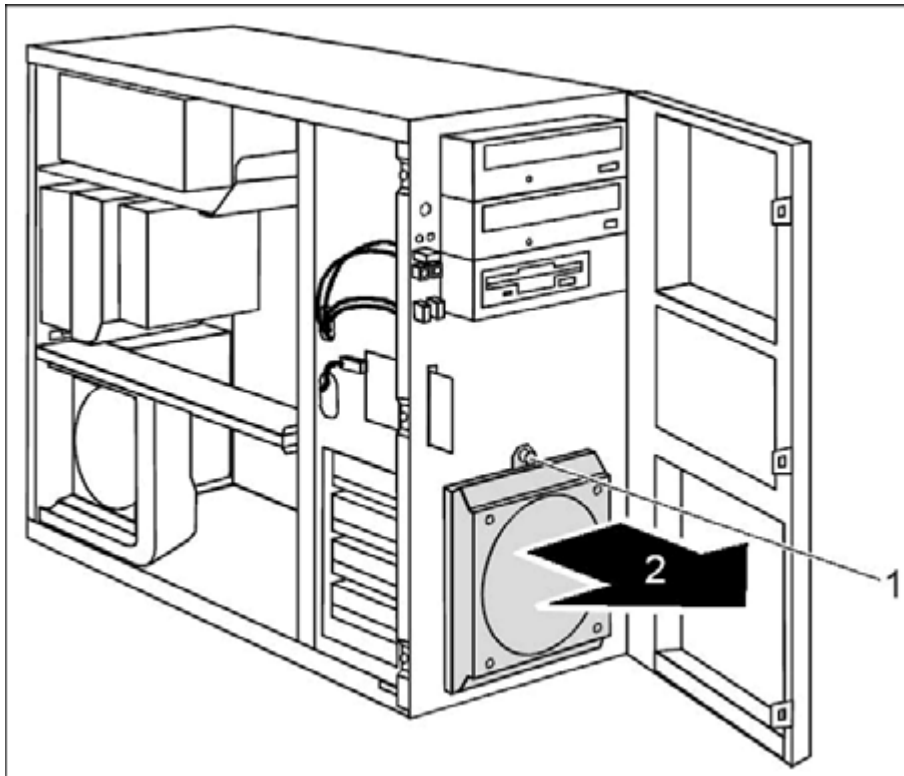


Fig. 5: Removing front fan

- Detach the green expansion rivet (1).
- Ensure that the green expansion rivet (1) is not pulled off the fan.
- Tilt the fan toward the front in the direction of the arrow (2) and remove it from the housing.
- Disconnect the fan cable (refer to [\(Fig. 4 / p. 12\)](#), “Front fan”).
- Pull the fan cable out of the housing.
- Remove the fan.

Securing the front fan on the fan bracket

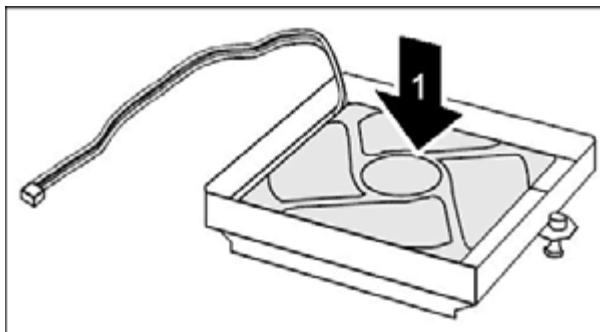


Fig. 6: Fan bracket

- Lay the fan onto the fan bracket in the direction of the arrow (1).

NOTE

When doing so, ensure the proper rotation and airflow direction.

Look around the label of the fan as shown in (Fig. 7 / p. 14).



Fig. 7: Air flow of the front fan

Press the fan firmly into the fan bracket until it is felt to engage.

Installing the front fan

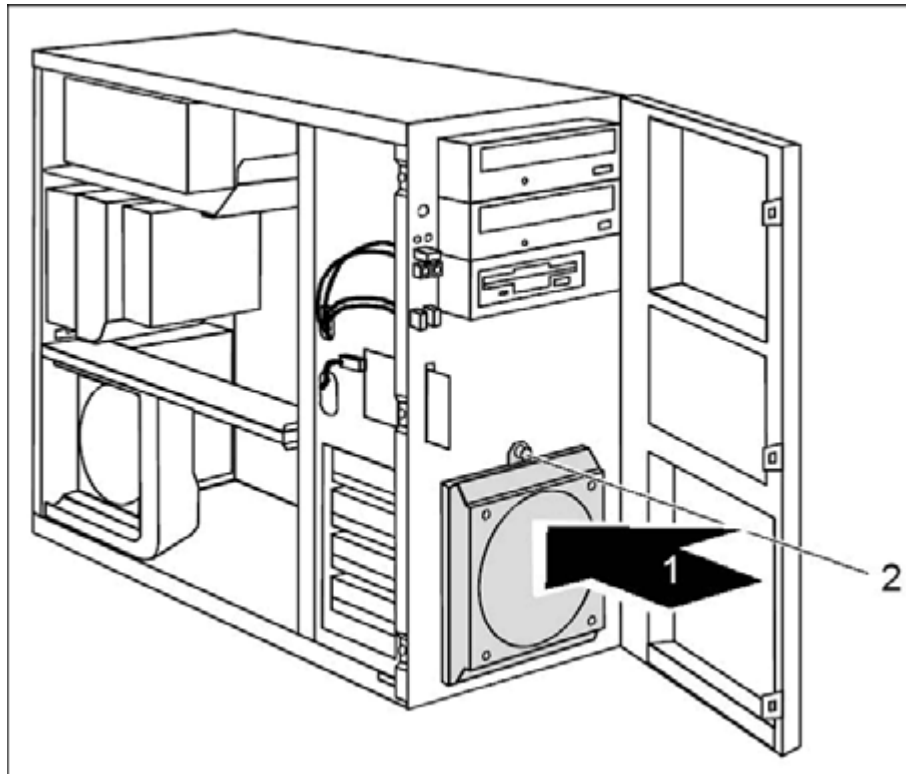


Fig. 8: Installing the front fan

- Connect the fan cable (refer to (Fig. 4 / p. 12), "Front fan").
- Insert the fan into the housing in the direction of the arrow (1) so that the lower edge of the fan is flush with the bottom of the device.

NOTE

When doing so, make sure not to pinch the power supply cable.

- Slightly pull back the green expansion rivet (2) until the tip of the expansion rivet is no longer spread.
- Press the green expansion rivet (2) into the hole on the housing.
- Secure the fan by pressing the green expansion rivet (2) into the housing until the tip of the expansion rivet is spread again.
- Close the front panel.
- Close the housing.

Side fan

You will need to remove the side fan inside the housing in order to install boards or upgrade the main memory.

Removing the side fan

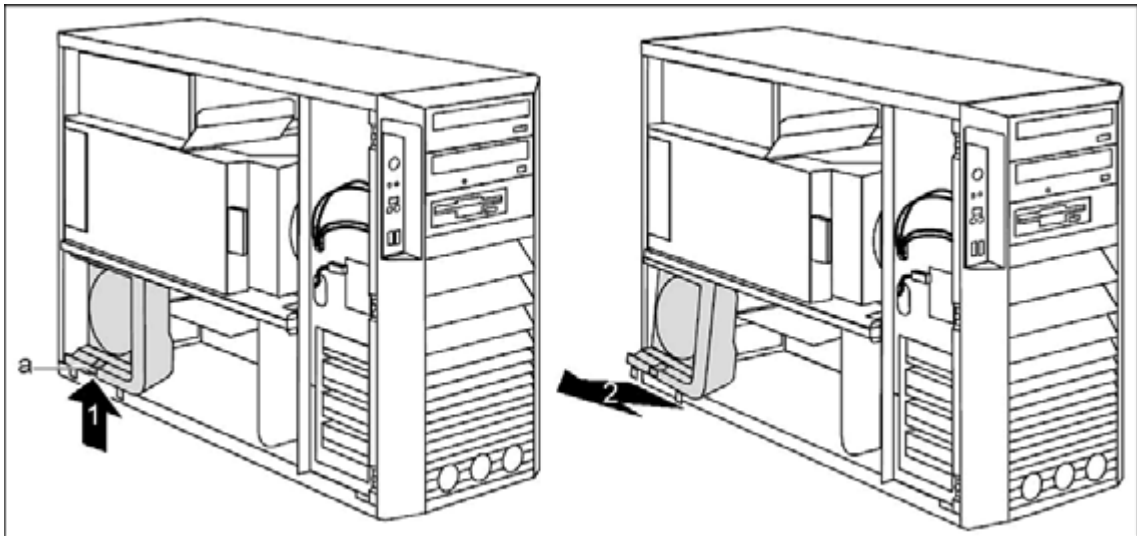


Fig. 9: Removing the side fan

- Open the housing (for better access remove [Crosspiece / p. 21](#)).
- Release the side fan by pressing the locking hook (a) in the direction of the arrow (1).
- Pull the bottom edge of the side fan in the direction of the arrow (2) out of the housing.
- Disconnect the cable of the fan from the mainboard (refer to [Fig. 4 / p. 12](#), "Side fan").
- Remove the side fan from the housing.

Installing the side fan

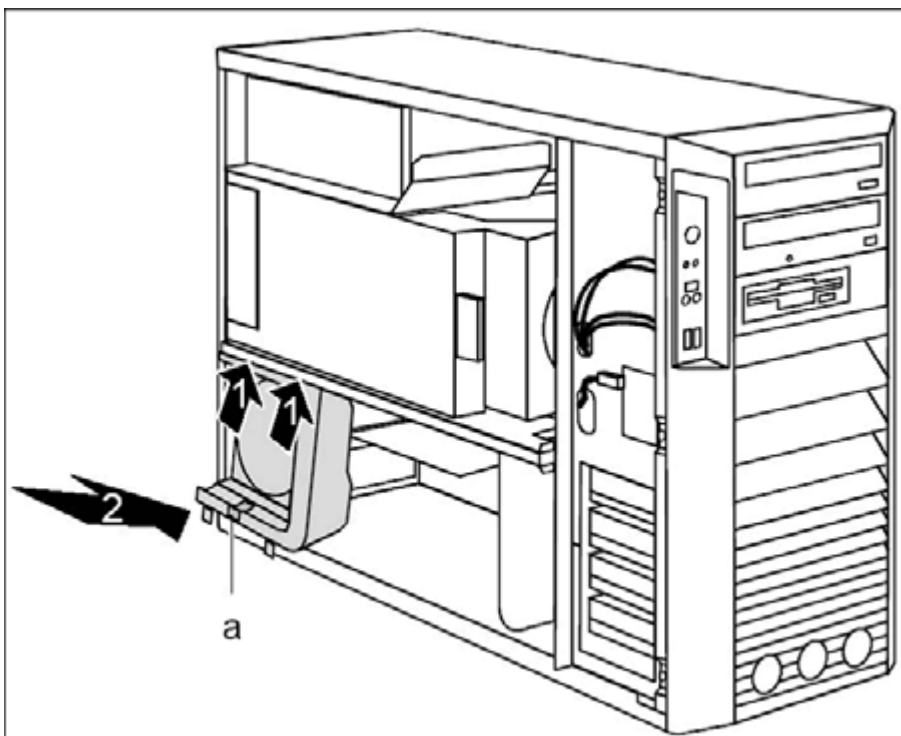


Fig. 10: Installing the side fan

- Connect the fan cable to the corresponding plug contact on the mainboard (refer to [\(Fig. 4 / p. 12\)](#), “Side fan”).
- Position the side fan at a slight angle in the direction of the arrow (1) and hook the fan lugs into the notches on the crosspiece.

NOTE

When doing so, ensure the proper rotation and airflow direction.

Look around the label of the fan as shown in [\(Fig. 11 / p. 17\)](#).

- Push the side fan into the housing in the direction of the arrow (2).



Fig. 11: Air flow of the side fan

- Lock the side fan by pressing the locking hook (a) downward.
- Close the housing.

Ventilation duct, CPU fan and rear fan

Removing the ventilation duct

Before you can upgrade the main memory or the processor, or before you can replace the processor or the lithium battery, you must remove the ventilation duct.

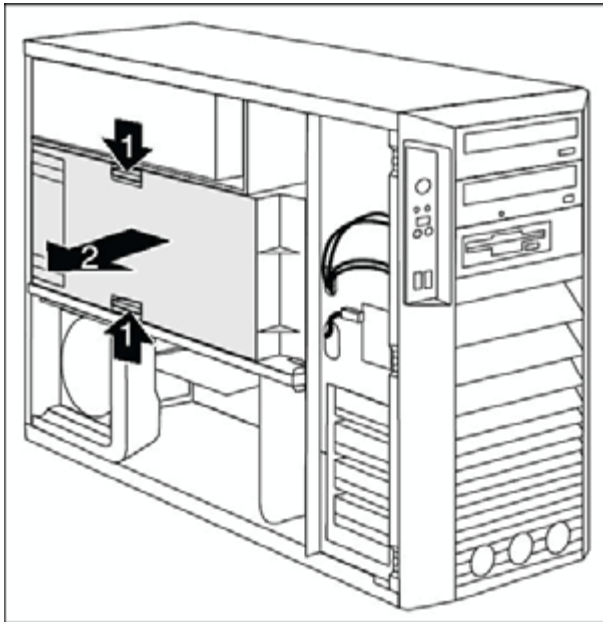


Fig. 12: Removing the ventilation duct

- Open the housing.
- Release the ventilation duct by pressing the locking hooks in the direction of the arrow (1).
- Grasp the ventilation duct by the cutouts and pull it in the direction of the arrow (2) out from the housing.

NOTE

When removing the ventilation duct, be careful not to damage the processor cooler(s) on the mainboard. If necessary, disconnect the cable to the mainboard.

Removing the CPU fan

- Disconnect the cable of the fan from the mainboard and take note of where it was plugged in (refer to (Fig. 4 / p. 12), “CPU fan”).
- Release the fan by pressing the locking hook. Remove the fan.

Removing the rear fan

- Disconnect the cable of the fan from the mainboard and take note of where it was plugged in (refer to (Fig. 4 / p. 12), "Rear fan").



Fig. 13: Locking hook of the rear fan

Release the fan by pressing the locking hook. Remove the fan.

Installing the CPU fan

- Connect the fan cable to the mainboard at the connection point you noted before (refer to (Fig. 4 / p. 12), "CPU fan").
- Mount the fan in the ventilation duct.

Installing the rear fan

- Connect the fan cable to the mainboard at the connection point you noted before (refer to (Fig. 4 / p. 12), "Rear fan").

- Mount the fan in the housing.



Fig. 14: Air flow of the rear fan

Installing the ventilation duct

NOTE

When routing the fan cable, make sure that it is not kinked or pinched.

When fitting the ventilation duct, be careful not to damage the processor cooler(s) on the mainboard.

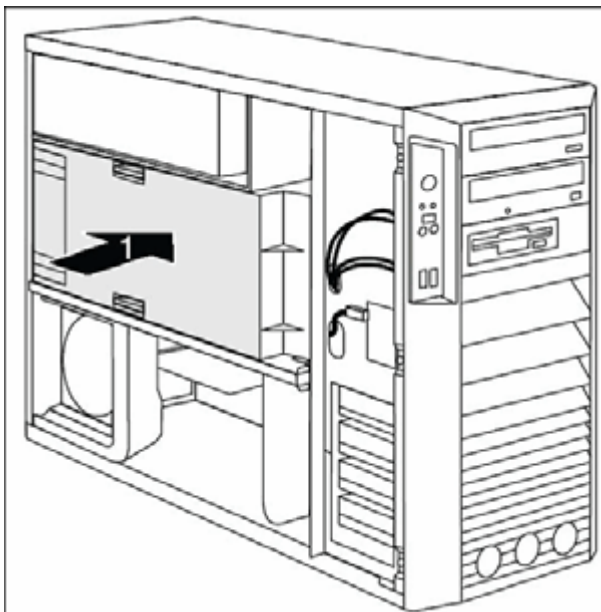


Fig. 15: Installing the ventilation duct

- Slide the ventilation duct in the direction of the arrow (1) into the housing until the locking hooks noticeably engage.
- Close the housing.

Crosspiece

Removing the crosspiece

Before replacing the main memory, you will need to remove the crosspiece.

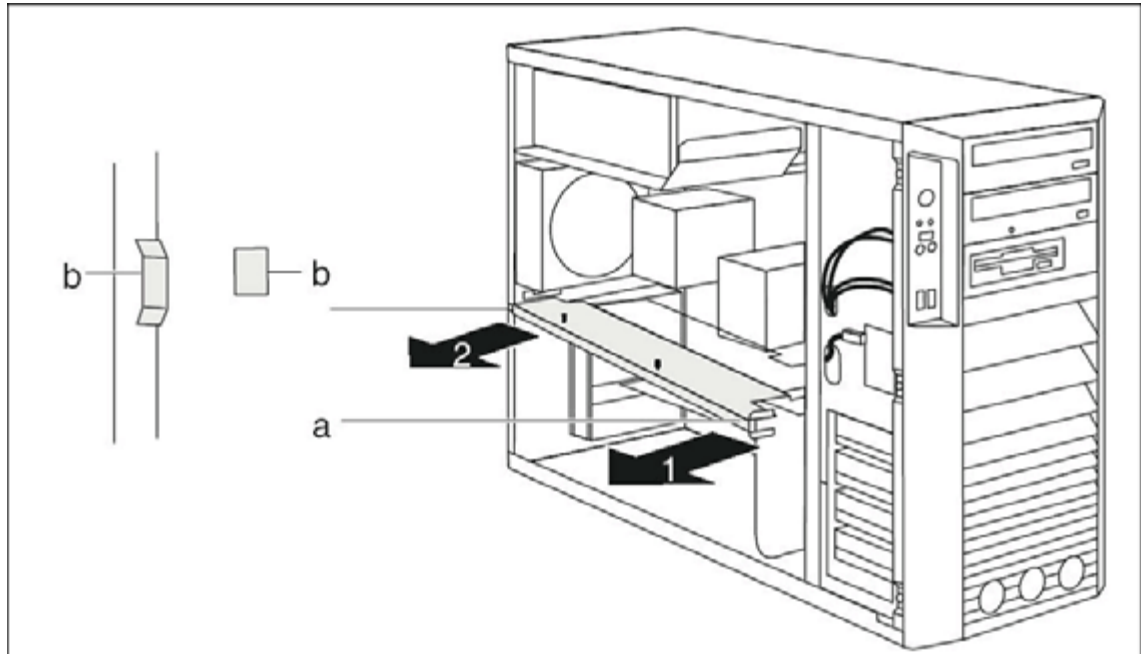


Fig. 16: Removing the crosspiece

- Open the housing.
- Remove the ventilation duct.
- Remove the side fan.
- Grasp the crosspiece by the lug (a) and unhook it from the brackets in the housing.
- Pull the crosspiece slightly out of the housing in the direction of the arrow (1).
- Detach the crosspiece from the brackets (b) on the back of the housing.
- Remove the crosspiece from the housing in the direction of the arrow (2).

Installing the crosspiece

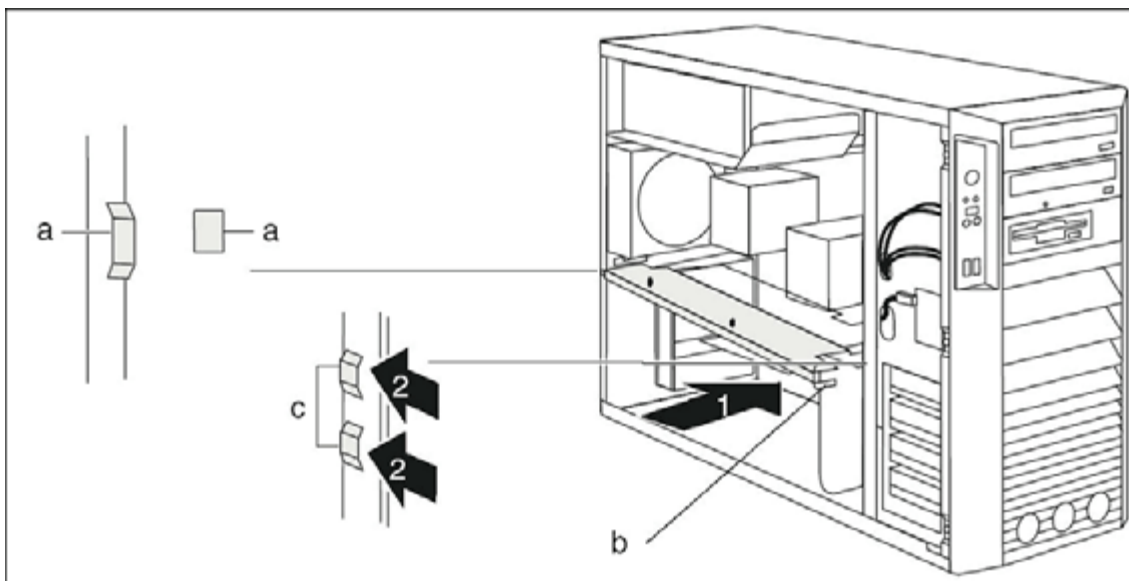


Fig. 17: Installing the crosspiece

- Insert the crosspiece into its brackets (a) on the back of the housing at a slight angle.

NOTE

Make sure that the Plexiglas plate of the crosspiece is not pinched between the memory slots.

- Press the crosspiece into the housing in the direction of the arrow (1) until the lugs (b) of the crosspiece hook into the brackets (c) in the housing.
- Install the side fan.
- Install the ventilation duct.
- Close the housing.

Power supply (R640 / R650)

Preparatory work steps

- Open the housing.
- Remove the ventilation duct.
- Remove the crosspiece.

NOTE

The power supply unit “PC/WE 1000W” only fits mechanically to the R650 host.

Changing the power supply



Fig. 18: Power Supply (1).jpg

- Make notes about the power connectors (mainboard [refer to [\(Fig. 20 / p. 25\)](#)], drives and graphics board) and the ducting of the power supply, and disconnect all of them.
- Remove the 4 screws (arrows) on the back of the power supply.
- (R650 remove the screw at the side from the power supply).
- Pull the power supply out from the case laterally.

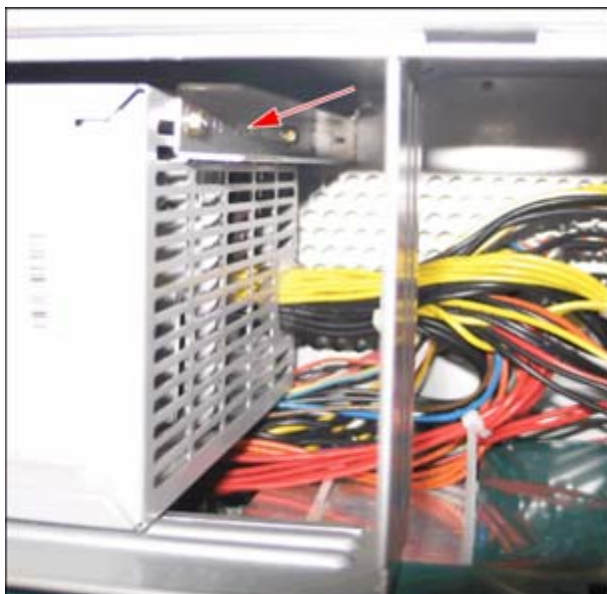


Fig. 19: Pillar

- (R640 Remove the pillar (2 screws) and mount it on the new power supply).
- Move the new power supply into the case and attach it with the 4 screws on the back.
- (R650 attach the screw at the side to the power supply).
- Connect all power connectors and attach the cables as they were before.

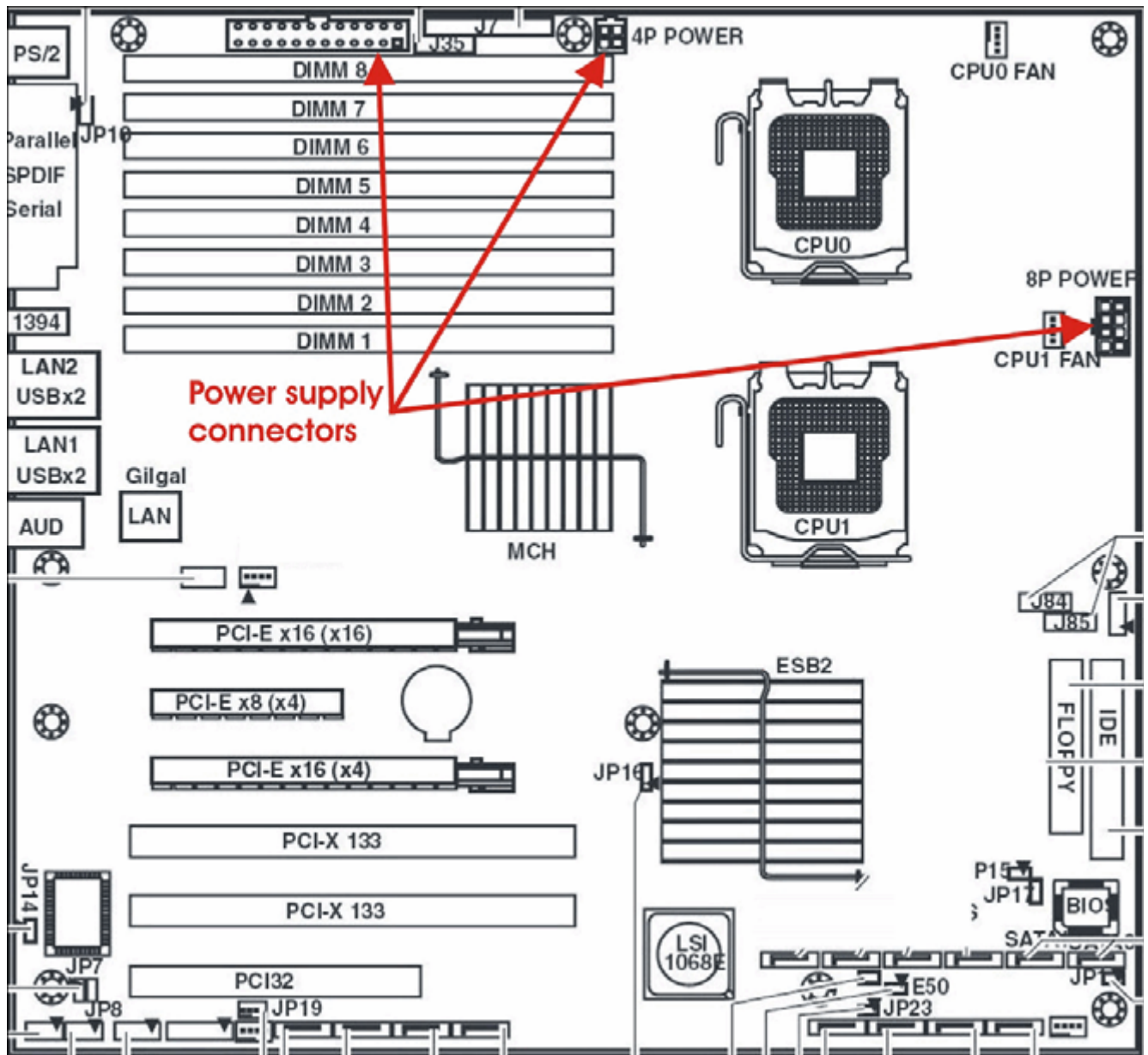


Fig. 20: Mainboard / power supply

Final work steps

- Install the crosspiece.
- Install the ventilation duct.
- Close the housing.

Battery

Changing the lithium battery

- Open the housing.

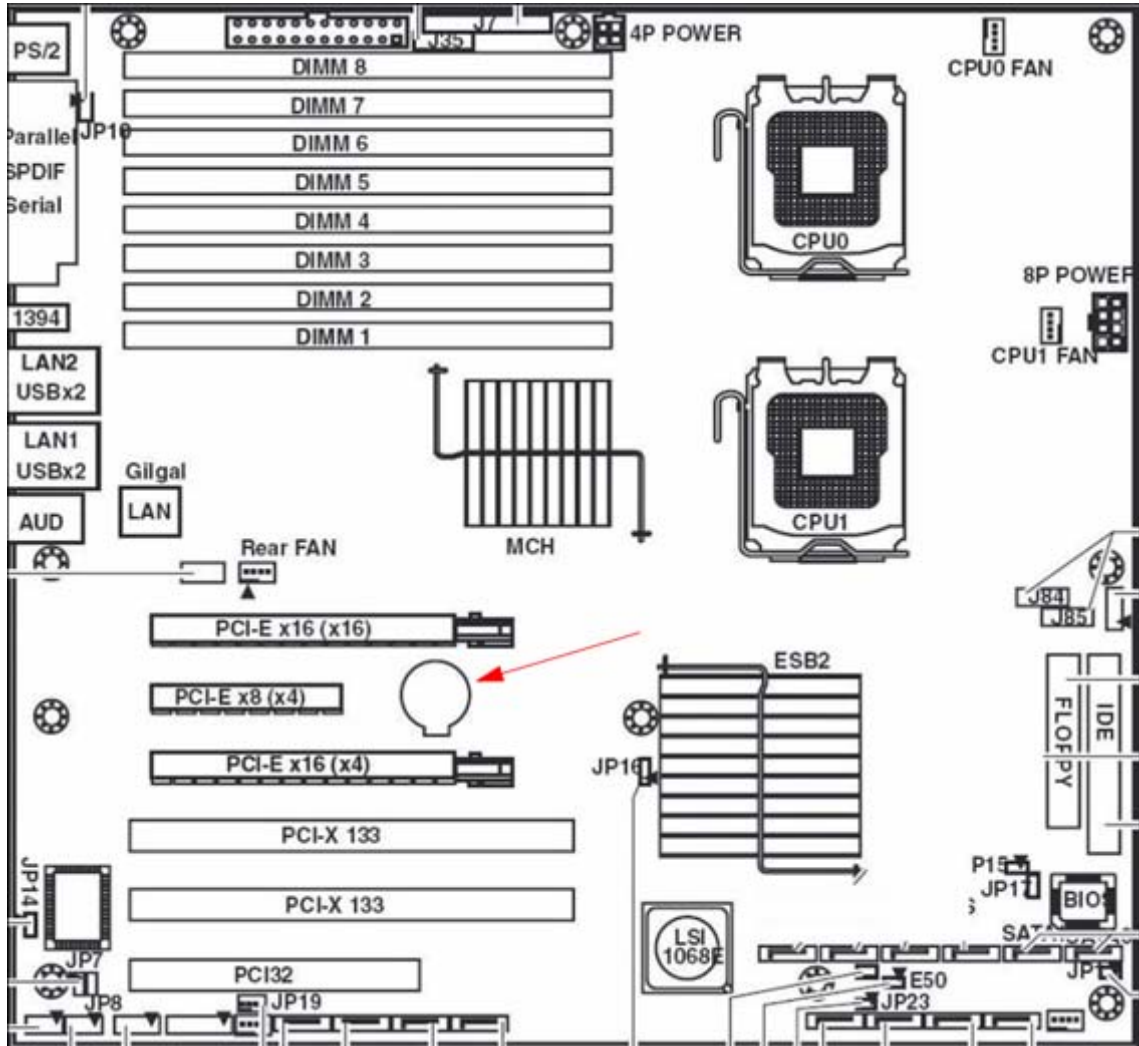


Fig. 21: Overview Mainboard D1808

In order to permanently save the system information, a lithium battery is installed to provide the CMOS memory with a current. A corresponding error message notifies the user when the charge is too low or the battery is dead. The lithium battery must then be replaced.



Incorrect replacement of the lithium battery may lead to a risk of explosion.

The lithium battery may be replaced only with an identical battery or with a type recommended by the manufacturer.

Do not throw lithium batteries into household waste. They must be disposed of in accordance with local regulations concerning hazardous waste.

Ensure that you observe the correct polarity when replacing the lithium battery. The plus pole must be on top.

There are several different lithium battery holder designs; all of them function in the same way.

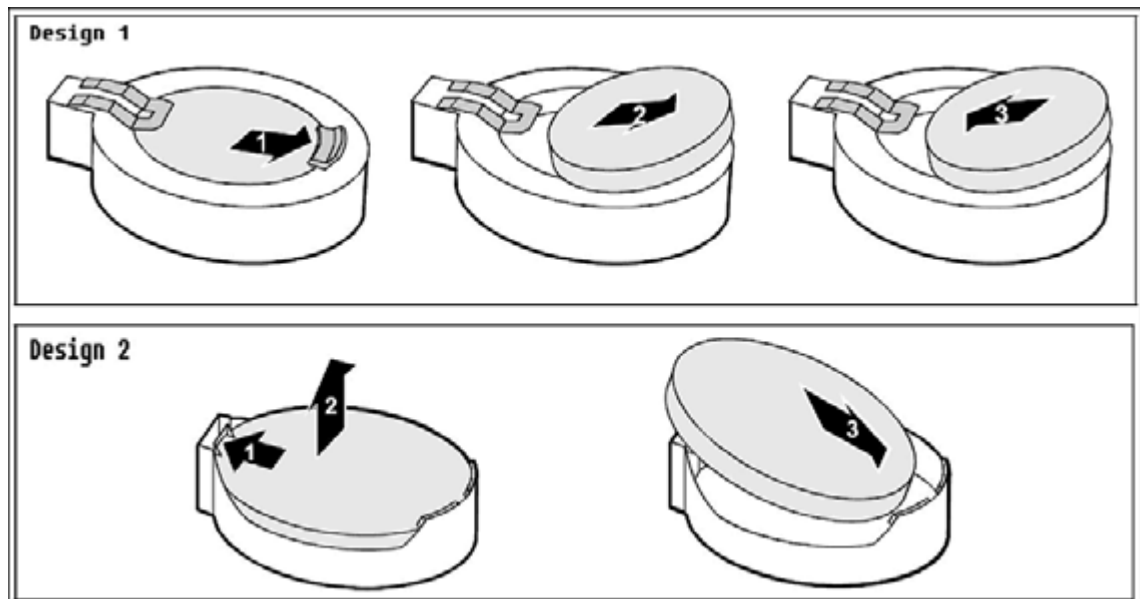


Fig. 22: Changing the lithium battery

- Press the catch in the direction of the arrow (1).
 - ⇒ The battery jumps out of the holder slightly.
- Remove the battery (2).
- Push the new lithium battery of the identical type into the holder and press it downward until it engages (3).

Restoring BIOS settings (necessary after battery replacement)

NOTE

When the battery is dead, the BIOS is set to factory default.
--

Final work steps

- Close the housing.
- Perform BIOS setup according to TDOC Installation Software.
- Check that the system boots properly

In-Room MRC

Required time and tools

Time

- 120 min.

Tools

- Set of standard tools
- TDOC Startup of the In-Room MRC

Final checks

- Run the auto configuration program and perform a visual evaluation according to the start-up of the In-Room MRC, Refer to TDOC start-up of the In-Room MRC

Protective conductor test



In case of a fault, voltage could be present at the housing of components.

This may lead to severe injuries or death if the resistance of the ground wire is too high.

- ⇒ **The protective conductor resistance must be $\leq 200 \text{ m}\Omega$. Measured against the LPD protective ground terminal, if a component replacement may affect the ground wire replacement.**

For more detailed information, refer to “Protective conductor measurement” in the binder: “Technical Documents”; register: Safety ; chapter: MR-specific safety information.

General

Safety information



Adhere to ESD regulations and use the appropriate tool set: anti-static wristband and grounded tabletop plate pad (TDK equipment Type 8005 from 3M Deutschland GmbH).

FRU overview

FRU	Time	Instructions	Final checks	Tools
Imager	70 min.		Perform “Image Orientation“ measurement. Service Software, Quality Assurance, General Quality Assurance, Image Orientation Perform a measurement, view the images produced.	Set of standard tools
HD container (Only valid for Quad Opteron)	45 min.		Perform a measurement, view the images produced.	Set of standard tools

CPU or HD container (High end)

The following information applies to replacement of the high end imager or the hard disk container:

Required time and tools

Time

- 70 min.

Tools

- Set of standard tools

Preliminary

- Shutdown and switch off the system.

Disassembly

- Replace the CPU High End (1) in the ACC.
- Replace the hard disk module (2) in the ACC.

NOTE

Ensure that the cables remains intact (at the back of the ACC).

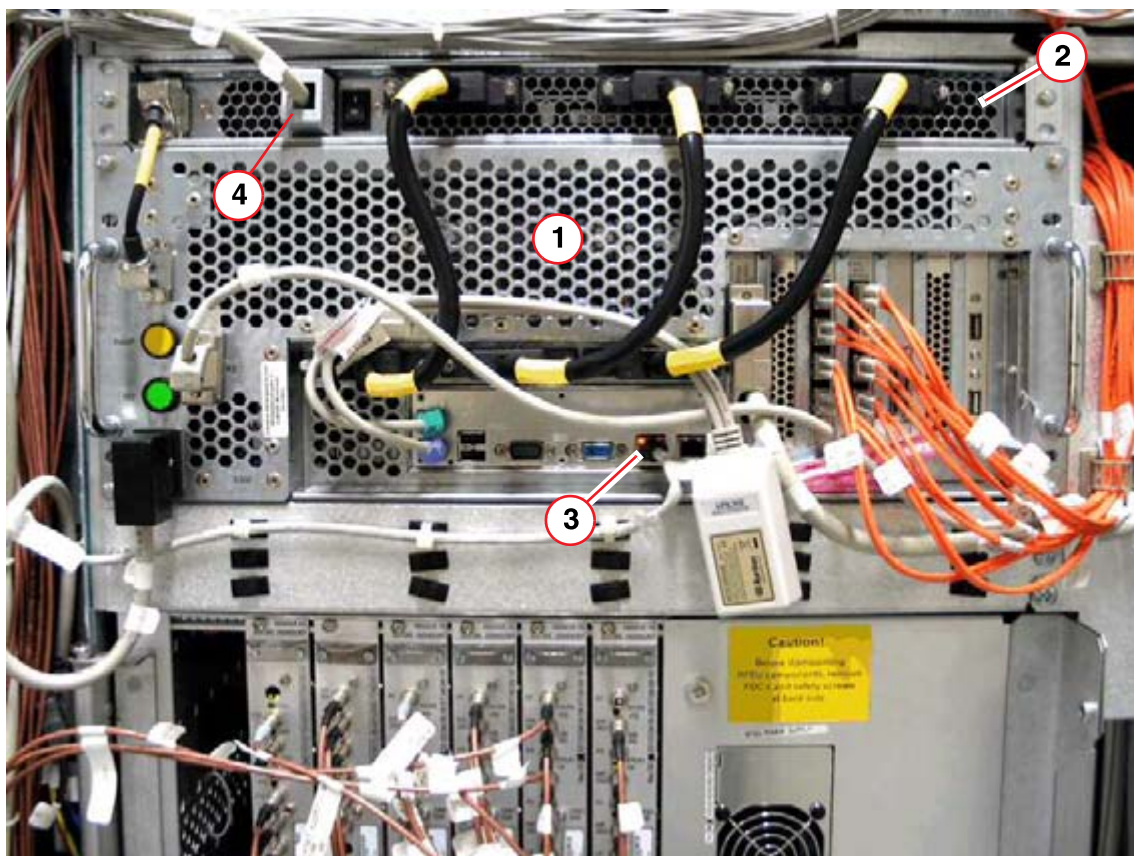


Fig. 23: ACC with high-end imager

- Pos. 1 CPU High End
Pos. 2 Hard hard disk module
Pos. 3 Ethernet connection
Pos. 4 PWR cable W3835

Assembling

- Reconnect the cables for the imager and the HD container
- Re-assemble the screws for the Imager / HD container to the cabinet

NOTE

Ensure that the cables remains intact (at the back of the ACC).

Final checks

- Switch on the system
- Start the Local Service, select **Quality Assurance**, Perform the “Quality Assurance” step Image Orientation
- Perform a measurement, view the images produced

Imager (low end)

The imager is heavy (approx. 20 kg).

Required time and tools

Time

- 70 min.

Tools

- Set of standard tools

Preliminary

- Shutdown and switch off the system.

Disassembly

NOTE

Ensure that the cables remains intact (at the back of the ACC).

- Disconnect the main power plug, network drop cable, fiber optic links, data receiver cable, and power remote switch cable.
- Remove the 4x Allen screws (within/below the side handles), pull/slide the “Imager” forward with care.
- Check that the number of PCI-RX boards is the same in the new “Imager” as the original “Imager”.

 Adhere to ESD regulations and use the appropriate tool set: anti-static wristband and grounded tabletop pad (TDK equipment Type 8005 from 3M Deutschland GmbH).

If the number of the PCI-RX boards differ between the old and the new “imager”, move the original PCI-RX boards to the new “imager”.

Assembling

- Insert the new “Imager”, mount the 4x Allen screws (within/below the side handles).
- Plug in all connectors/fiber fiber optic links at the front of the Imager
- Re-assemble the screws for the Imager container to the cabinet

NOTE

Ensure that the cables remains intact (at the back of the ACC).

Final checks

- Switch on the system
- Start the Local Service, select **Quality Assurance**, Perform the “Quality Assurance” step Image Orientation
- Perform a measurement, view the images produced

General information

NOTE

The monitor must be adjusted only if the monitor was not tuned to the PC at the factory, the option remote monitor will be installed or if the parts named below are replaced. The setting should be performed as described in the section Replacement of Parts.

Replacement of Parts:

- Monitor
- Graphics card
- PC

NOTE

This description does not apply to the In-room MRC.

NOTE

The monitor only needs to be checked (and adjusted as necessary) if at least one of the parts listed below is replaced.

Replacement parts:

- Monitor
- Graphics card
- PC

NOTE

Valid for Germany only.

If the monitor of an MR system is intended for medical reporting (as specified by the customer and labeled as such by the installation personnel), a special acceptance test is required in accordance with DIN V 6868-57. Based on the documentation listed below:

1. 'Acceptance test for image display devices' and
2. 'Acceptance and Constancy Test Record', an additional measurement and protocol will need to be approved by an authorized person (certified or TÜV).

Tools and time required

Time

approx. 60 min.

NOTE

Depending on the installed options (e.g. remote monitor, In-room MRC) and the accessibility to the cables, the exchange time can be increased up to 120 min.

Tools

- Screwdriver (Torx used for SIEMENS monitor; Philips used for CONRAC monitor)
- Spotmeter SMfit ACT, p/n: 77 52 848

Preliminary

- Switch off the monitor.

Disassembly

- Disconnect all cables.
- Remove the old monitor.

Assembly

- Install the new monitor.
- Reconnect all cables, if required. The wiring information can be found in the "Video cabling" section of the Start-up.

Final steps**NOTE**

Prior to evaluation, the TFT display must be in operation for at least 20 minutes (warm-up).

- Switch on the monitor.
Allow the monitor to warm up (20 min), and then perform the following steps:
 - Visually inspect/evaluate the TFT monitor ([Visually inspecting/evaluating the TFT monitor / p. 77](#))
- If the visual inspection reveals any problems, perform the adjustment work steps listed below.

Safety

- For your own safety and the safety of third parties, and to protect the MR system and its parts from damage, adhere to the safety information provided in “General Safety Notes” and “MR-specific Safety Information” in “Technical Documents”.
- Prior to beginning work, read the specific information in the manual regarding:
 - General safety information for service personnel,
 - Restrictions regarding ferromagnetic materials,
 - Safety information for working with the magnet at field,
 - Information for working with heavy loads,
 - The protective conductor test prior to switching on the system and following repairs.

Digital TFT monitor DSC 1904-D

Connections

General

The power supply and signal connections are located in back underneath a cover of the color flat panel display.

NOTE

Check that the cables are already positioned when you receive the display. The following steps are necessary only if an analog video input signal will be used at the monitor, e.g. usage as a remote monitor via the analog video signal line.

Switch off the monitor and PC before disconnecting or connecting the power or signal cables.

Display stand (removing)

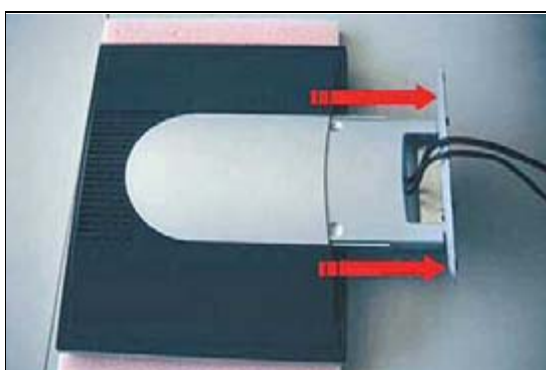


Fig. 24: Stand remove step 1

- Extend the display stand to the fully elongated position.

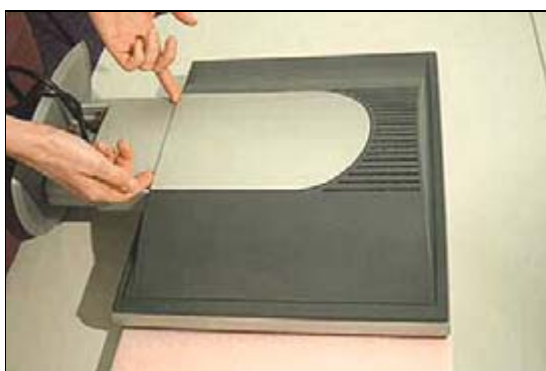


Fig. 25: Stand remove step 2

- Grasp the cover at the specified positions and remove it from the display.

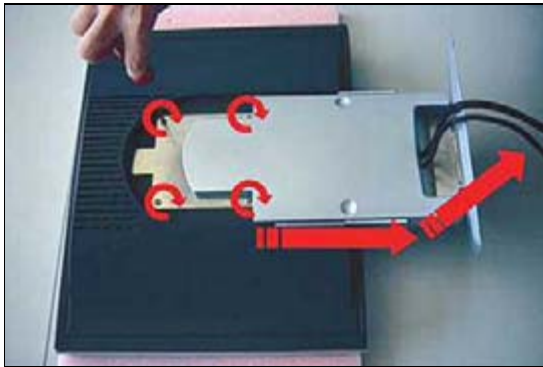


Fig. 26: Stand remove step 3

- Remove the four screws with an M4 Torx screwdriver (one turn suffices). Pull the stand downward and remove it.

Cable (removing)

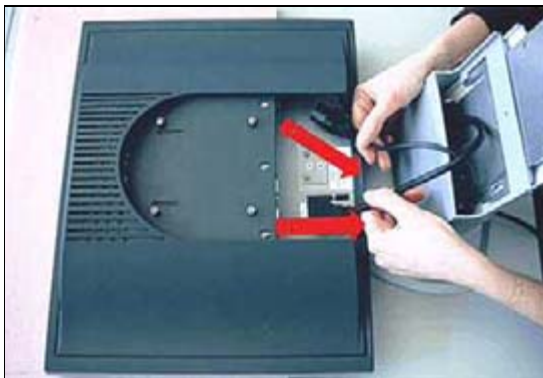


Fig. 27: Cable remove step 1

- Disconnect the cables from the display



Fig. 28: Cable remove step 2

- Remove the cables by leading them through the stand's cable duct

Cable (attaching)

Fig. 29: Cable connecting

- Lead the end of the cables through the stand's cable duct.
Connect the cables into the appropriate socket of the connecting panel of the monitor.
- For more information, refer to the **Connection panel**.

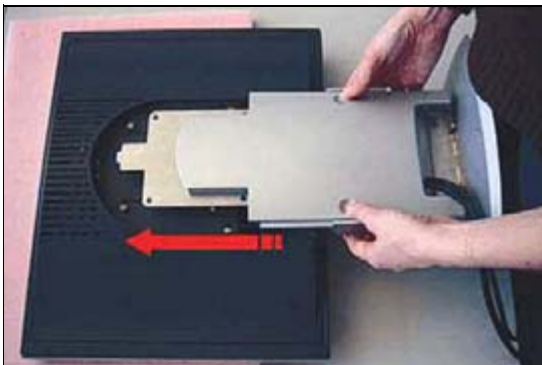
Display stand (attaching)

Fig. 30: Stand attaching

- Align the stand with the provided openings (screw holes) and tighten the screws



Fig. 31: Cover attaching

- Attach the cover

Installation information

- Only screened or compatible cables are permitted for the signal connections.
- All connectors should be protected against being pulled off, either by using a screw or latching (as far as possible).

Connection panel

The connection panel for the signals and power supply is located in back of the flat panel display underneath a cover (see Display Stand (removing)).

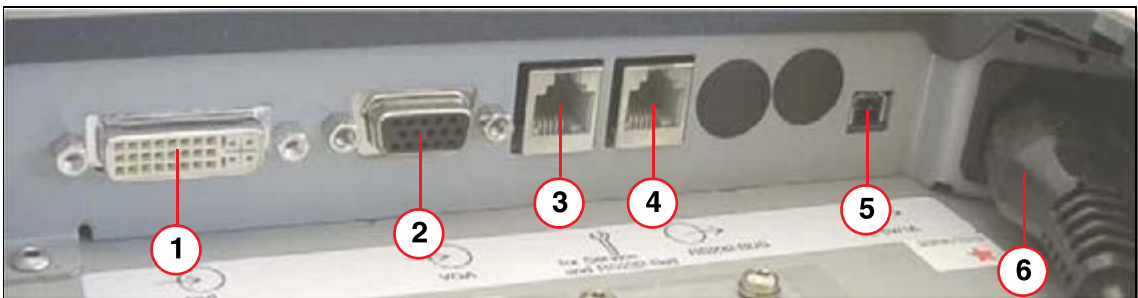


Fig. 32: Connection panel of the color flat panel display DSC 1904-D

- Pos. 1 Analog and digital inputs DVI connection
- Pos. 2 Analog input: 15-pin Sub-D connection
- Pos. 3 RS232 serial interface. RJ11 socket, male version
- Pos. 4 RS232 serial interface. RJ11 socket, male version
- Pos. 5 5 V/1 A power supply for external devices
- Pos. 6 Power supply connection

Adjustments

Introduction

Hints

- **Contrast:**
Use the Contrast parameter to adjust the perceptibility of the gray scale in the bright section.
- **Brightness:**
Use the Brightness parameter to adjust the perceptibility of the gray scale in the dark section.
- **Backlight:**
Use the Backlight parameter to adjust the maximum white level (**100% field**).

NOTE

If any parameter values have been changed via the OSD menu, wait at least thirty seconds before reading the display value. The adjustment requires a certain amount of time to reach the desired state.

NOTE

For the adjustment of TFT monitors, use spotmeter SMFit Act only.

TFT monitor 1904-D**NOTE**

Only applicable for TFT DSC 1904-D in combination with a CELSIUS R630 or newer PC hardware.

- The monitor will be delivered with connected power cord and digital video signal cable (DVI cable). The delivered and connected DVI cable transmits digital video signals only. If the monitor is used with a digital video input signal, no individual adjustment to the graphics card used or the PC is necessary.
- If the monitor is used with an analog video input signal via an additional video signal cable, the monitor has to be adjusted individually to the graphics card used or to the PC.
- A laptop is not required for adjustment. All adjustments can be performed in an on-screen menu, accessible using the pushbuttons (at the front of frame).
- The TFT monitor has an additional serial interface (6-contact mini DIN socket) for extended services. The mini DIN socket is visible from the front, at the bottom right behind the front panel. Specific instructions will be provided, if required.
- If the digital monitor is used in combination with an analog monitor, and if a signal splitting is to be arranged, refer to ([Video cabling for digital monitors / p. 68](#)).
- **Power switch:**
The TFT monitor has a power switch; as seen from the front, it is located at the bottom left behind the front panel.

NOTE

It is not necessary to open the monitor housing for service work.

A defective monitor must be replaced completely.



Fig. 33: TFT DSC 1904-D

- The DSC 1904-D (model designation) is a high-resolution 19" color TFT monitor (48 cm).
- The DSC 1904 is shipped with base and connection cables (Part no. 86 75 402).
- OSD operation is performed using the pushbuttons located at the bottom right of the front panel.

Monitor adjustment with the OSD menu

Button functions of the TFT monitor, DSC1904-D

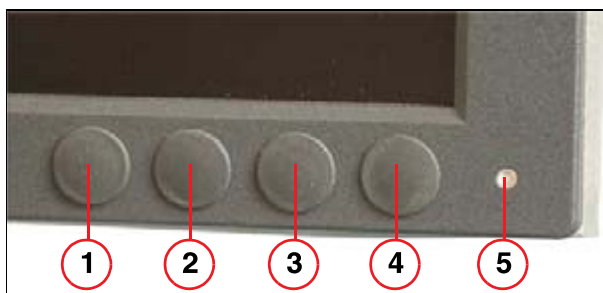


Fig. 34: DSC 1904-D (push buttons)

Pos. 1	Menu
Pos. 2	Up ↑
Pos. 3	Down ↓
Pos. 4	Set
Pos. 5	Power LED

- The status of the TFT monitor is indicated by the power LED. ON / OFF / Standby.
- The front buttons are designed without a "designation".
- All required adjustments are made using the pushbuttons located on the front frame.

The pushbuttons are blocked when shipped (new system).

- Unlock or lock them as follows:

Press 1x "SET", then 3x ↑.

Press the buttons in rapid succession.

The OSD menu can then be selected using the Menu button.

NOTE

For safety reasons, the OSD menu is locked by the manufacturer. Only authorized personnel may unlock it. After adjustments are performed, it must be locked again.

OSD menu

Tab. 1 Button functions in the OSD menu

Button(s)	Action
Menu	Selection of the OSD / menu item
up ↑	Opens the menu item, a submenu can be recognized by the fact that the menu title is highlighted. Sets the value up or to the right.
down ↓	Sets the value down or to the left.
Set	Closes the menu item (press one level higher 1x).

Tab. 2 Locking/unlocking the OSD menu

Function	Action
OSD menu locking or unlocking	Press the Set button 1x, then quickly press the ↑ three times. To lock or unlock, the OSD menu has to be closed. If the OSD is locked, only the input signal can be switched with the Up ↑ and Down ↓ buttons.
Adjustment values Saving	Press the Set button as often as required until the "Undo" menu item appears. Any changes made are accepted by pressing "Accept changes". "New" settings can be rejected by pressing "Reject changes". Selection is made by pressing the "Menu" button. Select via the "Up" ↑ button.

Tab. 3 OSD menu

Menu	Brightness / Contrast	Brightness	Adjustable
		Contrast	Adjustable
		Backlight	Adjustable
		Color	Selection => 2
		Set user color	not enabled
	Position / Zoom	H-POSITION	Adjustable
		V-POSITION	Adjustable
		Zoom	Selection => Fill screen
	Picture source	Source selection	Not needed; selection is automatic
	Auto functions	Auto brightness contrast	Use only if the monitor has been adjusted incorrectly.
		Auto position phase frequency	Auto adjustment of size / position / phase and frequency.
		Execute selected auto functions	Start the selected auto functions.
	Language	German/English	Selection => English
	Other	Frequency/phase	Manual adjustment of frequency and phase.
		Sharpness	Selection => 3
		OSD settings	If required, the position and the "transparency" of the OSD menu can be set here.
		DPMS settings	Selection ON
		Status	Status display of the monitor (e.g. temp., operating hours.....)

Troubleshooting

General

- In the case of a malfunction, the TFT monitor is completely replaced.
- For replacement part numbers, refer to the SPC (Spare Parts Catalog).
- Every defective monitor must be returned with an exact description of the malfunction. Without a description of the malfunction, it is virtually impossible to find sporadic and/or temperature-related effects or even system-related causes.

Troubleshooting and solutions

Malfunction	Possible cause	Solution
TFT monitor displays no image, power LED is off.	Defective fuse. Power cable not plugged in or line power missing.	1. Fuse okay, line power present. 2. Replace the TFT monitor.
TFT monitor displays no image, power LED is on.	No video/synchronous signal (cable or video source). Inputs incorrectly plugged in with a new installation.	1. Video/synchronous signal present. 2. Replace the TFT monitor.
Unclear image, defects in vertical lines	Frequency and/or phase incorrectly set. Disruptions to the video signal.	1. For frequency / phase setting, refer to the Adjustment procedure . 2. Ensure that the video signal is not experiencing any interferences. 3. Replace the TFT monitor.
Other malfunctions (sporadic)	Plug-in connection(s) is/are loose.	1. Ensure that there are no "control problems" (line power/video). 2. Replace the TFT monitor.
Required contrast no longer achieved?	Video signal is too low.	1. The video signal must be at least 0.5 V, otherwise "full control" is no longer guaranteed. 2. A brightness/contrast adjustment using the Adjustment procedure does not improve matters 3. Replace the TFT monitor.

Digital TFT monitor DSC 1907-D

Connections

General

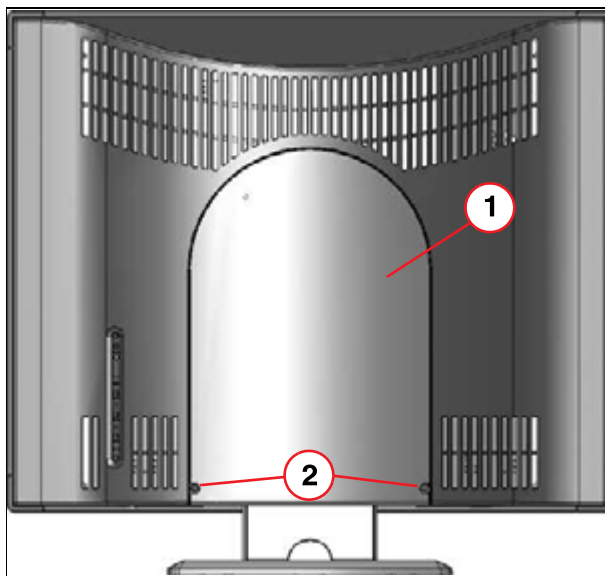
The power supply and signal connections are located in back underneath a cover of the color flat panel display.

NOTE

Check that the cables are already positioned when you receive the display. The following steps are necessary only if an analog video input signal will be used at the monitor, e.g. usage as a remote monitor via the analog video signal line.

Switch off the monitor and PC before disconnecting or connecting the power or signal cables.

Removing the display cover



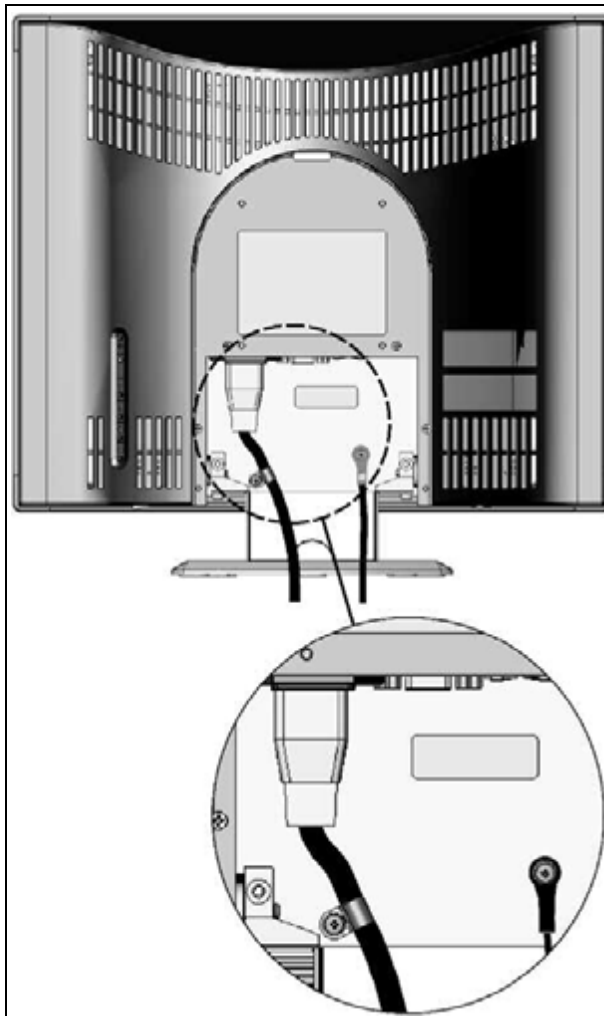
- Release the two cover screws with a Philips screwdriver and remove the cover.

Fig. 35: Remove cover

Pos. 1 Connection panel cover

Pos. 2 Philips screws

Attaching the cable



- Connect the cables into the appropriate socket of the monitor connection panel (install the power cable with the lead cleat at the monitor, then the video signal cable).
- For more information, refer to the **Connection panel**.

Fig. 36: Attaching the cable



- Strictly adhere to the assembly sequence if an additional ground connection has to be installed.

Fig. 37: Assembly sequence for ground connection

Installation information

- Only screened or compatible cables are permitted for the signal connections.
- All connectors should be protected against being pulled off, either by using a screw or latching (as far as possible).
- After the cable is connected, install the cover and tighten the screws.

Connection panel

The connection panel for the signals and power supply is located in back of the flat panel display underneath a cover (refer to Removing the display cover).

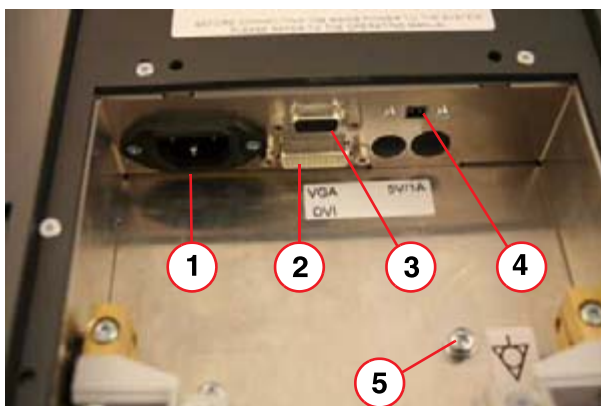


Fig. 38: Connection panel of monitor DSC 1907-D

- Pos. 1 Power supply connection
- Pos. 2 Analog and digital inputs for DVI connection
- Pos. 3 Analog input: 15-pin Sub-D connection
- Pos. 4 5 V power supply for external devices
- Pos. 5 Ground connector

Adjustments

Introduction

Hints

- **Contrast:**
Use the Contrast parameter to adjust the perceptibility of the gray scale in the bright section.
- **Brightness:**
Use the Brightness parameter to adjust the perceptibility of the gray scale in the dark section.
- **Backlight:**
Use the Backlight parameter to adjust the maximum white level (**100% field**).

NOTE

If any parameter values have been changed via the OSD menu, wait at least thirty seconds before reading the display value. The adjustment requires a certain amount of time to reach the desired state.

NOTE

For the adjustment of TFT monitors, use spotmeter SMFit Act only.

TFT monitor 1907-D

NOTE

Only applicable for TFT DSC 1907-D in combination with a CELSIUS R630 or newer PC hardware.

- The monitor will be delivered with connected power cord and digital video signal cable (DVI cable). The delivered and connected DVI cable transmits digital video signals only. If the monitor is used with a digital video input signal, no individual adjustment to the graphics card used or the PC is necessary.
- If the monitor is used with an analog video input signal via an additional video signal cable, the monitor has to be adjusted individually to the graphics card used or to the PC.
- A laptop is not required for adjustment. All adjustments can be performed in an on-screen menu, accessible using the pushbuttons (at the front of frame).
- The TFT monitor has an additional serial interface (6-contact mini DIN socket) for extended services. The mini DIN socket is visible from the front and located at the bottom left. Specific instructions will be provided, if required.
- If the digital monitor is used in combination with an analog monitor, and if a signal splitting is to be arranged, refer to ([Video cabling for digital monitors / p. 68](#)).
- **Power switch:**
The TFT monitor has a main power switch; visible from the front and located at the bottom right.

NOTE

It is not necessary to open the monitor housing for service work.

A defective monitor must be replaced completely.

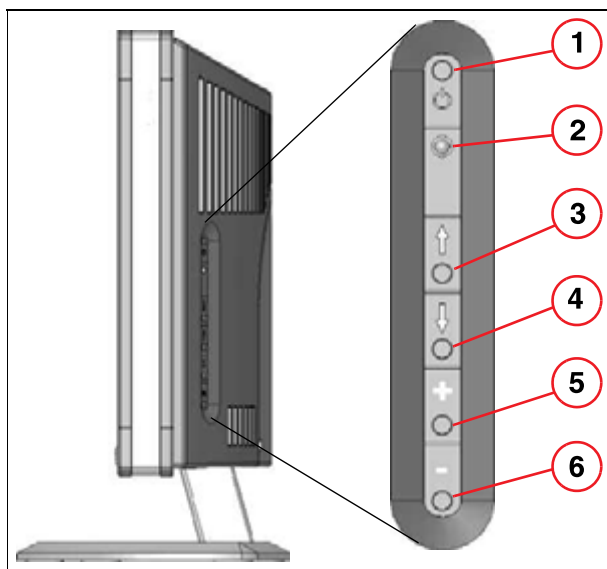


Fig. 39: TFT DSC 1907-D

- The DSC 1907-D (model designation) is a high-resolution 19" color TFT monitor (48 cm).
- The DSC 1907-D is shipped with base and connection cables (Part No. 10293002).
- OSD operation is performed using the pushbuttons; visible from the front, located on the right behind the front panel.

Monitor adjustment with the OSD menu

Button functions on the TFT monitor, DSC 1907-D



- The status of the TFT monitor is indicated by the power LED. ON / OFF / Standby.
- All required adjustments are made using the pushbuttons; viewed from the front, located to the right on the rear side.
- The OSD menu is locked. To unlock press: "+" and "-" simultaneously until "Keypad unlocked" appears and afterwards the OSD main menu is displayed.
- All changes in the OSD menu are stored immediately.

Fig. 40: DSC 1907-D (push-buttons)

Pos. 1	Power / stand-by
Pos. 2	Operation LED
Pos. 3	Up ↑
Pos. 4	Down ↓
Pos. 5	"+" button
Pos. 6	"-" button

NOTE

For safety reasons, the OSD menu is locked by the manufacturer. Only authorized personnel may unlock it. When closing the OSD menu, it will lock automatically.

OSD menu

Tab. 4 Button functions in the OSD menu

Button(s)	Action
Power	Standby mode on / off (press button for about 2 seconds)
Operation LED	Green: Monitor is switched on and video signal available. Red: Standby mode on. Red flashing: No video signal available.
up ↑	Selects the previous menu item.
down ↓	Selects the next menu item.

Button(s)	Action
+	Open the next submenu. If “+” is displayed in a menu item, the next submenu can be opened. Increases the parameters (e.g. brightness) Select the next settings (e.g. sharpness).
-	If applied in the main menu the OSD menu will be closed. Decreases the parameters (e.g. brightness). Select the next settings (e.g. sharpness). Return to previous menu: Select the menu item “←” and press the button “-”. If the OSD menu is not activated (displayed) the signal inputs (RGB analog, RGB digital) can be toggled.

Tab. 5 Locking/unlocking the OSD menu

Function	Action
OSD menu locking or unlocking	Press the “+” and “-” buttons simultaneously (for about 2 seconds) until “Keypad unlocked” appears. The OSD main menu is then displayed on the screen. When closing the OSD menu, it will lock automatically.
Adjustment values Saving	All changes are stored immediately. Via: Options -> More Options -> Reset to defaults, the specified factory settings can be reactivated.

Tab. 6 OSD menu

Menu	Picture	Brightness	Adjustable
		Contrast	Adjustable
		H / V position	for analog RGB only
		Phase / frequency	for analog RGB only
		Scaling	fill all: The horizontal and vertical resolution is scaled to the screen size. one to one: The picture is displayed in its original resolution on the screen. fill aspect ratio: The picture is scaled up to the full screen.
	Display	Backlight	Adjustable
		Gamma	Sets the correction of the gamma curve. The color values will be provided with the specified factor and sent to the display. 1: Alignment according to DICOM. 2: Alignment according to CIE. Fix: Linear.
		Color matrix Color temperature: User red, User green, User blue	Enables or disables the color matrix. Color temperature: Select a defined value. User: Individual color temperature through setting of the colors red, green, blue.
	Options	OSD h pos	Moves the OSD menu in a horizontal direction.
		OSD v pos	Moves the OSD menu in a vertical direction.
		OSD time out	Set the time after which you wish the unused OSD menu to fade out automatically.
		Background	Sets the OSD background: opaque or transparent.
		Language	Sets the OSD menu language (English).
		More options	See menu: More options.

Menu	Info	Temperature	Manual adjustment of frequency and phase.
		Software version	Selection => 3
		Model	If required, the position and the "transparency" of the OSD menu can be set here.
		Serial number	Selection ON
		Current video signal	Resolution / vertical frequency / mode

Tab. 7 OSD menu: More options

More options	DPMS	Displays the power management system. On: If no video signal is available, the monitor switches to standby mode. Off: If no video signal is available, the monitor does not switch to standby mode.
	Source scan	On: Automatic scan of all inputs. Off: Manual scan of the inputs via "-" button, if no OSD menu is displayed.
	Border color	Set the border color, only visible with the setting "scaling one to one". Settings: red, green, blue or black
	Test pattern	Test pattern for setting the display.
	Reset to defaults	Reset all parameters and functions to the factory / specified delivery defaults.

Troubleshooting

General

- In the case of a malfunction, the TFT monitor is completely replaced.
- For replacement part numbers, refer to the SPC (Spare Parts Catalog).
- Every defective monitor must be returned with an exact description of the malfunction. Without a description of the malfunction, it is virtually impossible to find sporadic and/or temperature-related effects or even system-related causes.

Troubleshooting and solutions

Malfunction	Possible cause	Solution
TFT monitor displays no image, power LED is off.	Defective fuse. Power cable not plugged in or line power missing.	1. Fuse okay, line power present. 2. Replace the TFT monitor.
TFT monitor displays no image, power LED is on.	No video/synchronous signal (cable or video source). Inputs incorrectly plugged in with a new installation.	1. Video/synchronous signal present. 2. Replace the TFT monitor.
Unclear image, defects in vertical lines	Frequency and/or phase incorrectly set (for analog RGB only). Disruptions to the video signal.	1. For frequency / phase setting, refer to the Adjustment procedure . 2. Ensure that the video signal is not experiencing any interferences. 3. Replace the TFT monitor.
Other malfunctions (sporadic)	Plug-in connection(s) is/are loose.	1. Ensure that there are no "control problems" (line power/video). 2. Replace the TFT monitor.

Digital TFT monitor DSC 1907-D Rev02

Connections

General

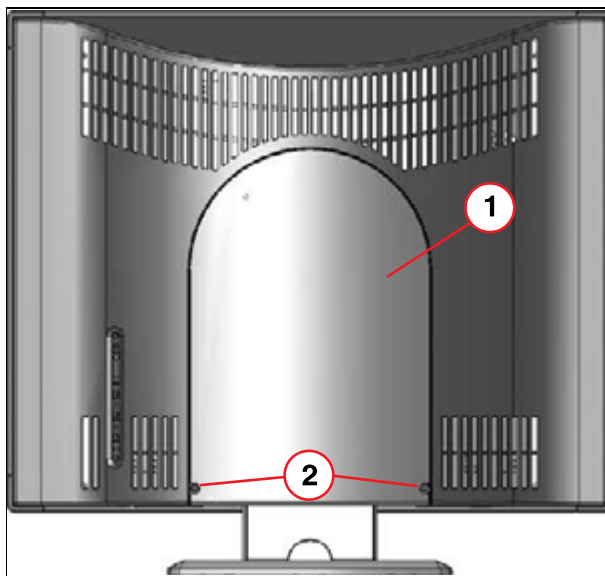
The power supply and signal connections are located at the back underneath a cover of the color flat panel display.

NOTE

Check that the cables are already positioned when you receive the display. The following steps are necessary only if an analog video input signal will be used at the monitor, e.g. usage as a remote monitor via the analog video signal line.

Switch off the monitor and PC before disconnecting or connecting the power or signal cables.

Removing the display cover



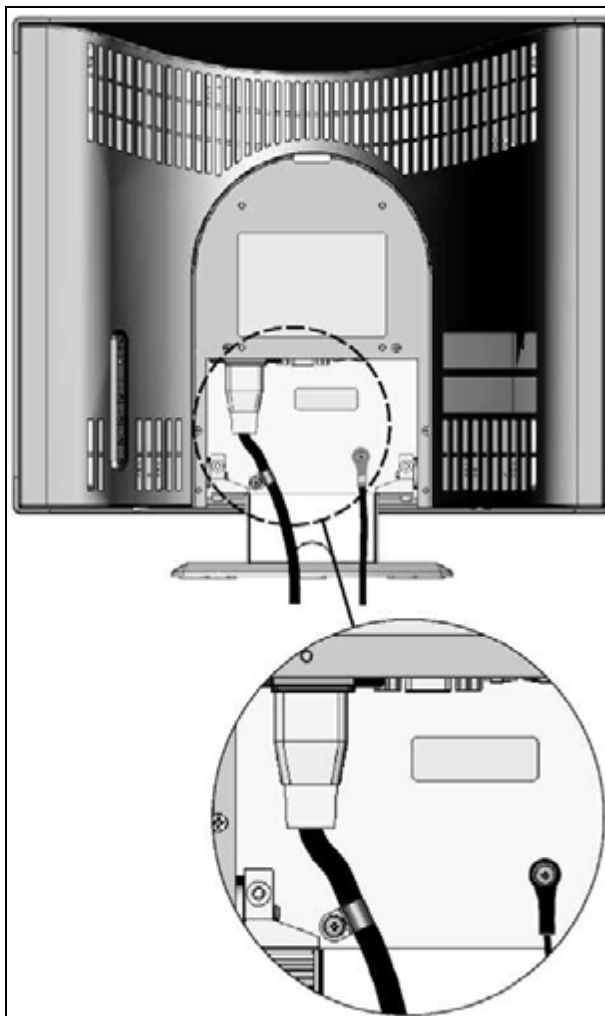
- Release the two cover screws with a Philips screwdriver and remove the cover.

Fig. 41: Remove cover

Pos. 1 Connection panel cover

Pos. 2 Philips screws

Attaching the cable



- Connect the cables into the appropriate socket of the monitor connection panel (install the power cable with the lead cleat at the monitor, then the video signal cable).
- For more information, refer to the **Connection panel**.

Fig. 42: Attaching the cable



- Strictly adhere to the assembly sequence if an additional ground connection has to be installed.

Fig. 43: Assembly sequence for ground connection

Installation information

- Only screened or compatible cables are permitted for the signal connections.
- All connectors should be protected against being pulled off, either by using a screw or latching (as far as possible).
- After the cable is connected, install the cover and tighten the screws.

Connection panel

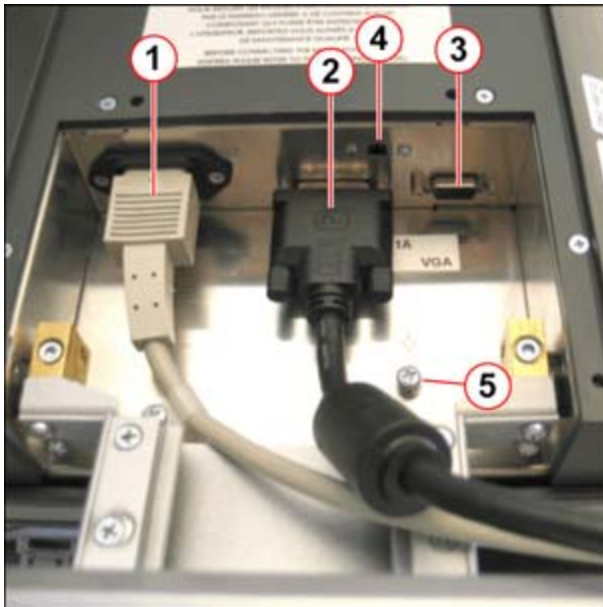


Fig. 44: Connection panel of the DSC 1907-D Rev.02 monitor

- Pos. 1 Power supply connection
- Pos. 2 Analog and digital inputs for DVI connection
- Pos. 3 Analog input: 15-pin Sub-D connection
- Pos. 4 5V power supply for external devices
- Pos. 5 Ground connector

The connection panel for the signals and power supply is located at the back of the flat panel display underneath a cover (refer to Removing the display cover).

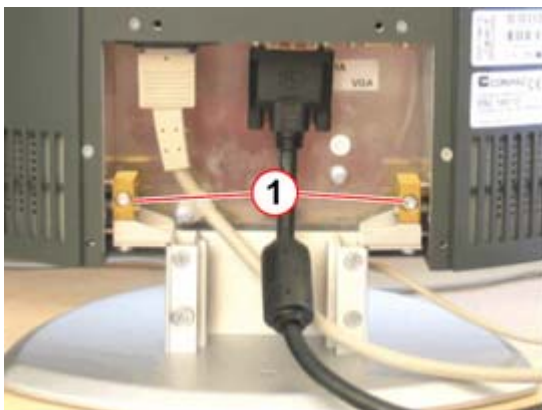


Fig. 45: Adjusting the tilting device (DSC 1907-D Rev.02)

- Pos. 1 Screws for the adjustment of the tilting device

- If the tilting device slips through or can not be moved, you can change the setting at the marked screws (1).

Adjustments

Introduction

Hints

- **Contrast:**
Use the Contrast parameter to adjust the perceptibility of the gray scale in the bright section.
- **Brightness:**
Use the Brightness parameter to adjust the perceptibility of the gray scale in the dark section.
- **Backlight:**
Use the Backlight parameter to adjust the maximum white level (**100% field**).

NOTE

If any parameter values have been changed via the OSD menu, wait at least thirty seconds before reading the display value. The adjustment requires a certain amount of time to reach the desired state.

NOTE

For the adjustment of TFT monitors, use spotmeter SMFit Act only.

TFT monitor 1907-D Rev02

NOTE

Only applicable for hosts with digital video signal transfer (e.g. CELSIUS R630 or later).

- The monitor will be delivered with connected power cord and digital video signal cable (DVI cable). The delivered and connected DVI cable transmits digital video signals only. If the monitor is used with a digital video input signal, no individual adjustment to the graphics card used or the PC is necessary.
- If the monitor is used with an analog video input signal via an additional video signal cable, the monitor has to be adjusted individually to the graphics card used or to the PC.
- A laptop is not required for adjustment. All adjustments can be performed in an on-screen menu, accessible using the pushbuttons (at the front of frame).
- The TFT monitor has an additional serial interface (6-contact mini DIN socket) for extended services. The mini DIN socket is visible from the front and located at the bottom left. Specific instructions will be provided, if required.
- If the digital monitor is used in combination with an analog monitor, and if a signal splitting is to be arranged, refer to [\(Video cabling for digital monitors / p. 68\)](#).
- **Power switch:**
The TFT monitor has a main power switch; visible from the front and located at the bottom right.

NOTE

It is not necessary to open the monitor housing for service work.

A defective monitor must be replaced completely.

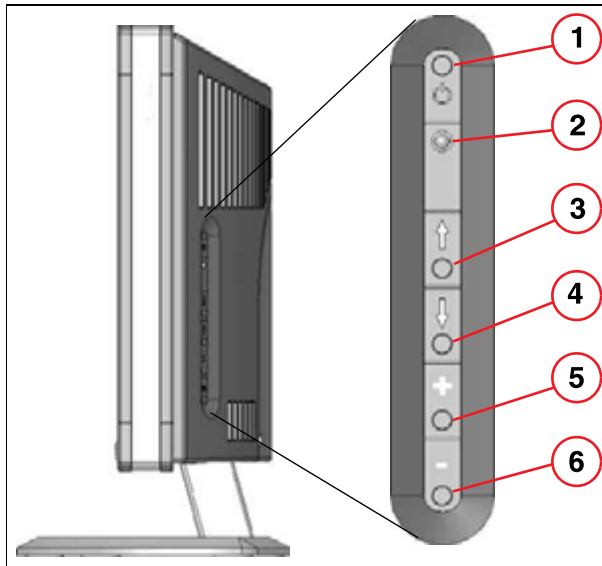


Fig. 46: TFT DSC 1907-D Rev.02

- The DSC 1907-D Rev02 (model designation) is a high-resolution 19" color TFT monitor (48 cm) without the Siemens logo on the front side.
- The DSC 1907-D Rev02 is shipped with base and connection cables (Part No. 10293002).
- OSD operation is performed using the pushbuttons; visible from the front, located on the right behind the front panel.

Monitor adjustment with the OSD menu

Button functions of the TFT monitor, DSC 1907-D Rv02



- The status of the TFT monitor is indicated by the power LED. ON / OFF / Standby.
- All required adjustments are made using the pushbuttons; viewed from the front, located to the right on the rear side.
- The OSD menu is locked. To unlock press: “+” and “-” simultaneously until the OSD main menu is displayed.
- All changes in the OSD menu are stored immediately.

Fig. 47: DSC 1907-D (push-buttons)

Pos. 1	Power / stand-by
Pos. 2	Operation LED
Pos. 3	Up ↑
Pos. 4	Down ↓
Pos. 5	“+” button
Pos. 6	“-” button

NOTE

For safety reasons, the OSD menu is locked by the manufacturer. Only authorized personnel may unlock it. When closing the OSD menu, it will lock automatically.

OSD menu

Tab. 8 Button functions in the OSD menu

Button(s)	Action
Power	Standby mode on / off (press button for around 2 seconds), back or exit function in the OSD menu.
Operation LED	Green: Monitor is switched on and video signal available. Red: Standby mode on. Red flashing: No video signal available.
up ↑	Selects the previous menu item.
down ↓	Selects the next menu item.

Button(s)	Action
+	Opens the selected menu. If “+” is displayed in a menu item, the next submenu can be opened. Increases the parameters (e.g. brightness) Selects the next settings (e.g. color balance).
-	Decreases the parameters (e.g. brightness). Selects the previous setting (e.g. color balance). Returns to previous menu.

Tab. 9 Locking/unlocking the OSD menu

Function	Action
OSD menu locking or unlocking	Press the “+” and “-” button simultaneously (for about 2 seconds) until the OSD main menu is displayed on-screen. When closing the OSD menu, it will lock automatically. No settings can be adapted without a video signal.
Adjustment values Saving	All changes are stored immediately. Via: Setup -> OSD -> Reset to defaults, the specified factory settings can be reactivated.

Tab. 10 OSD menu

Menu	Picture	Contrast	Adjustable
		Brightness	Adjustable
		H / V position	for analog RGB only
		Sharpness	Adjustable
		Color balance	Four options can be selected via - / + buttons: Normal, Warm, Cold and User. User: Individual color temperature through setting of the colors red, green, blue.
	Display	Backlight	Adjustable
		Gamma	Sets the correction of the gamma curve. The color values will be provided with the specified factor and sent to the display. Setting range: Off, 1: Alignment according to DICOM. 2: Alignment according to CIE. If 1 or 2 is chosen the backlight will be reset to the calibrated value.
		Zoom (DVI formats)	Aspect ratio: Displayed picture will be horizontally corrected. Fill all: The screen is filled in a horizontal direction. Auto Zoom: The current picture format triggers an automatic switch between 4:3 and 16:9. Panorama: non-linear scaling in horizontal and vertical direction (in center = original scaling, but more intense scaling toward the edges). 14:9 Top: Picture in 14:9 letterbox format with subtitle. 14:9: Zoom enlarged vertically (15% is cut off), the picture is reduced in size horizontally. 16:9 Top: Picture in 16:9 letterbox format with subtitle. 16:9: Picture is scaled horizontally, cut off at the top and bottom.

Menu	Setup/OSD	Time out	Sets the OSD display time duration from the last activation until the OSD display is closed.
		Transparency	The transparency can be switched on or off and adapted ranging from 1 to 10.
		Show logo	The appearance of the logo can be turned on or off during switch-on or in the case of a signal disruption.
		Restore default settings	Restores the default factory settings, selection using the “ + / - ” button, the “Yes” function and by pressing the power button.
		Language	The language of the OSD menu can be selected via “ + / - ”.
	Setup/Info	Current temperature	Displays the current temperature of the device.
		Software version	Version of the current installed software.
		Software revision	Revision of the current installed software.
		Model	Model name
		Serial number	Monitor serial number
	Setup/DPMS	DPMS	Displays the power management system. On: If no video signal is available, the monitor switches to standby mode. Off: If no video signal is available, the monitor does not switch to standby mode.
	Setup/source scan	Source scan	On: Automatic scan of all inputs. Off: Manually scans the inputs (approximately 2 seconds) via “-” button, if the OSD menu is not displayed.

Troubleshooting

General

- In the case of a malfunction, the TFT monitor is completely replaced.
- For replacement part numbers, refer to the SPC (Spare Parts Catalog).
- Every defective monitor must be returned with an exact description of the malfunction. Without a description of the malfunction, it is virtually impossible to find sporadic and/or temperature-related effects or even system-related causes.

Troubleshooting and solutions

Malfunction	Possible cause	Solution
TFT monitor displays no image, power LED is off.	Power cable not plugged in or line power missing.	1. Replace the TFT monitor.
TFT monitor displays no image, power LED is on.	No video/synchronous signal (cable or video source). Inputs incorrectly plugged in with a new installation.	1. Check if video/synchronous signal is present. 2. Replace video signal cable. 3. Replace the TFT monitor.
Unclear image, defects in vertical lines	Frequency and/or phase incorrectly set (for analog RGB only). Disruptions to the video signal.	1. Replace video signal cable. 2. Ensure that the video signal is not experiencing any interferences. 3. Replace the TFT monitor.
Other malfunctions (sporadic)	Plug-in connection(s) is/are loose.	1. Ensure that there are no "control problems" (line power/video). 2. Replace the TFT monitor.

Video cabling for digital monitors

With additional options and digital/analog signal transmission

NOTE

If options such as Remote Color Monitor or in-Room MRC have been installed on your system, the cabling must be different.

NOTE

A Video Multiplexer (also known as video distribution box) is needed to spread the analog signal when using additional options with the Remote Color Monitor (refer to [\(Fig. 54 / p. 74\)](#))

- Shut down the MR system
- Option Remote Monitor (only), see [\(Fig. 49 / p. 70\)](#)
 - Analog video signal transmission via the Video Multiplexer (video distribution box) to the Remote Monitor (complete analog video signal transmission);
 - Digital video signal transmission to digital input of the MRC TFT
- Option In-Room MRC and Remote Monitor, see [\(Fig. 50 / p. 71\)](#)
 - Analog video signal transmission via Video Multiplexer (video distribution box) to the Remote Monitor (complete analog video signal transmission);
 - Digital video signal transmission via Interconnection Box to digital MRC TFT monitor and In-Room MRC (digital video signal transmission)
- Option In-Room MRC [\(Fig. 51 / p. 72\)](#)
 - Digital video signal transmission via Interconnection Box to digital MRC TFT monitor and In-Room MRC (digital video signal transmission).
- Route the video cables according [\(Fig. 48 / p. 69\)](#) to [\(Fig. 51 / p. 72\)](#) of the used monitors.
- Connect the monitor to the MRC power distributor using the power cords.

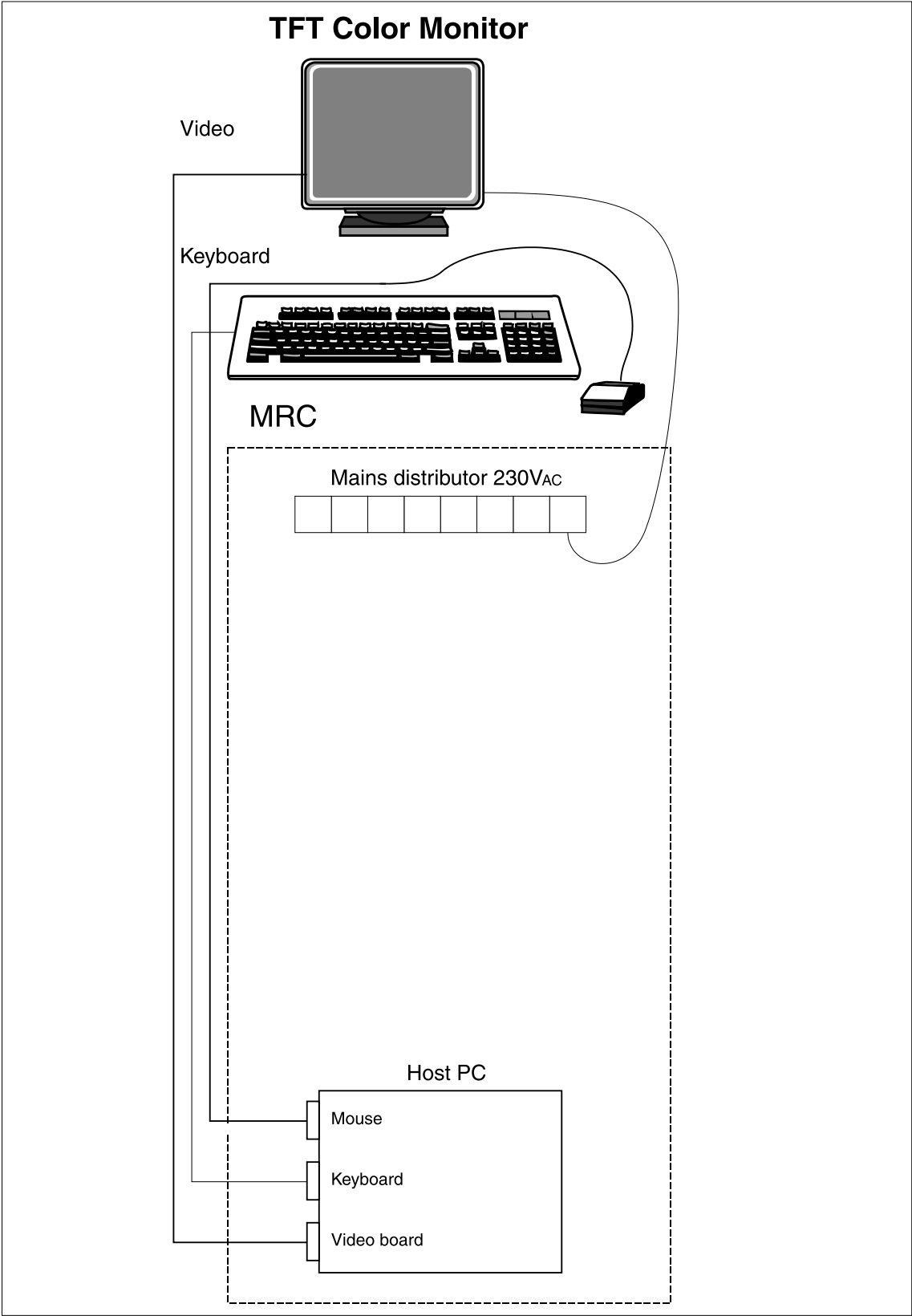


Fig. 48: Cabling for the TFT color monitor

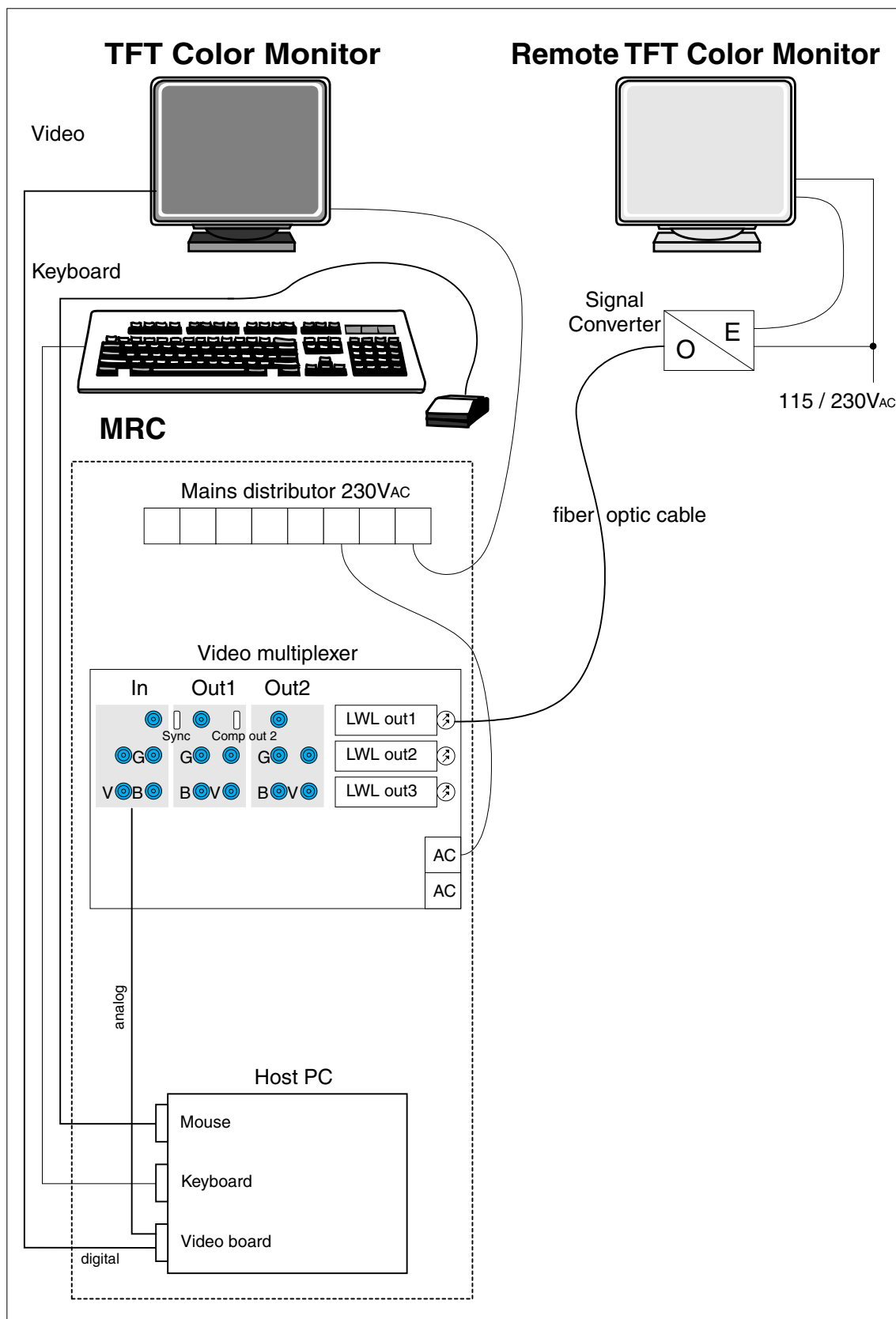


Fig. 49: Cabling for the TFT remote monitor

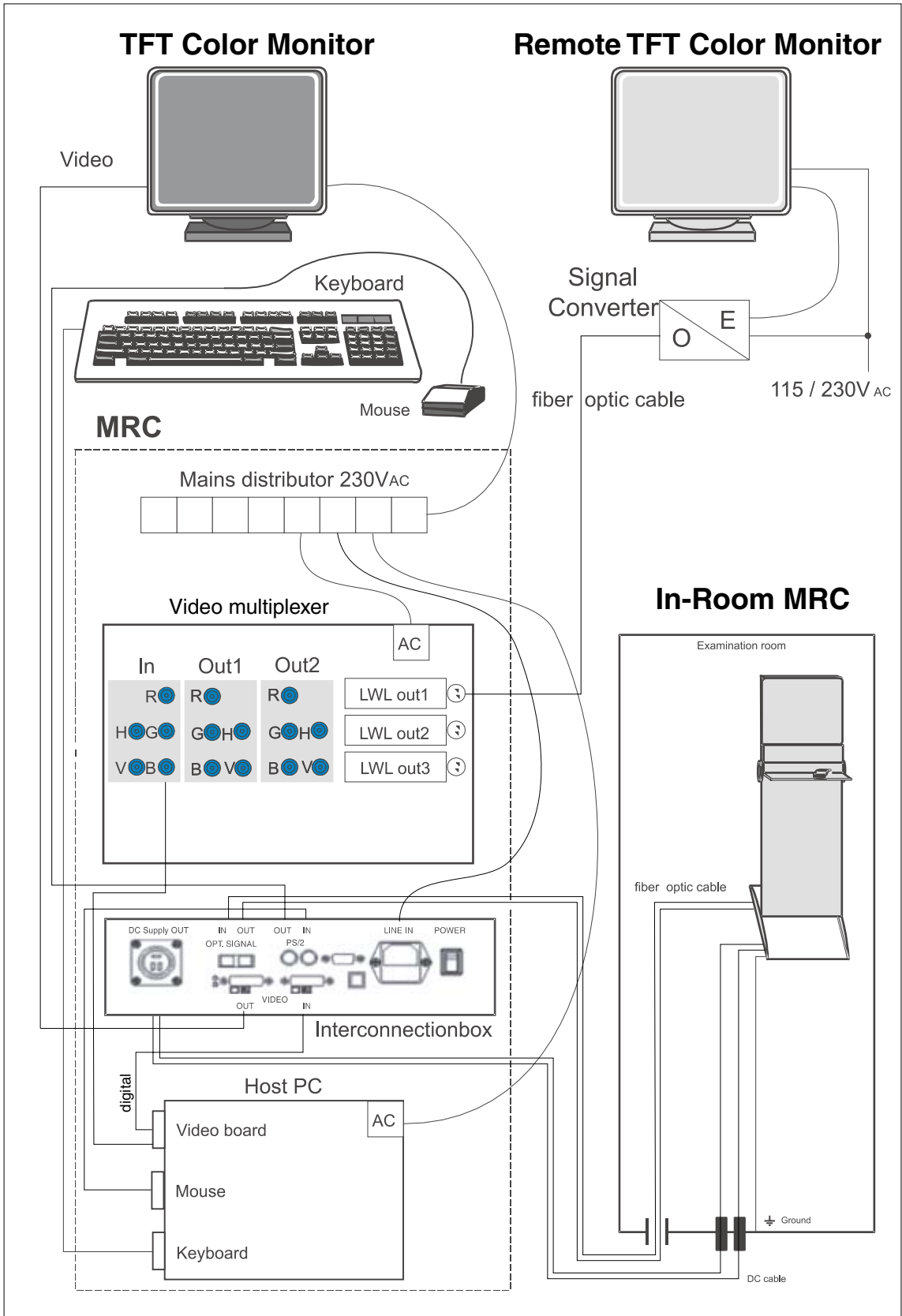


Fig. 50: Cabling for TFT remote monitor and In-room MRC monitor

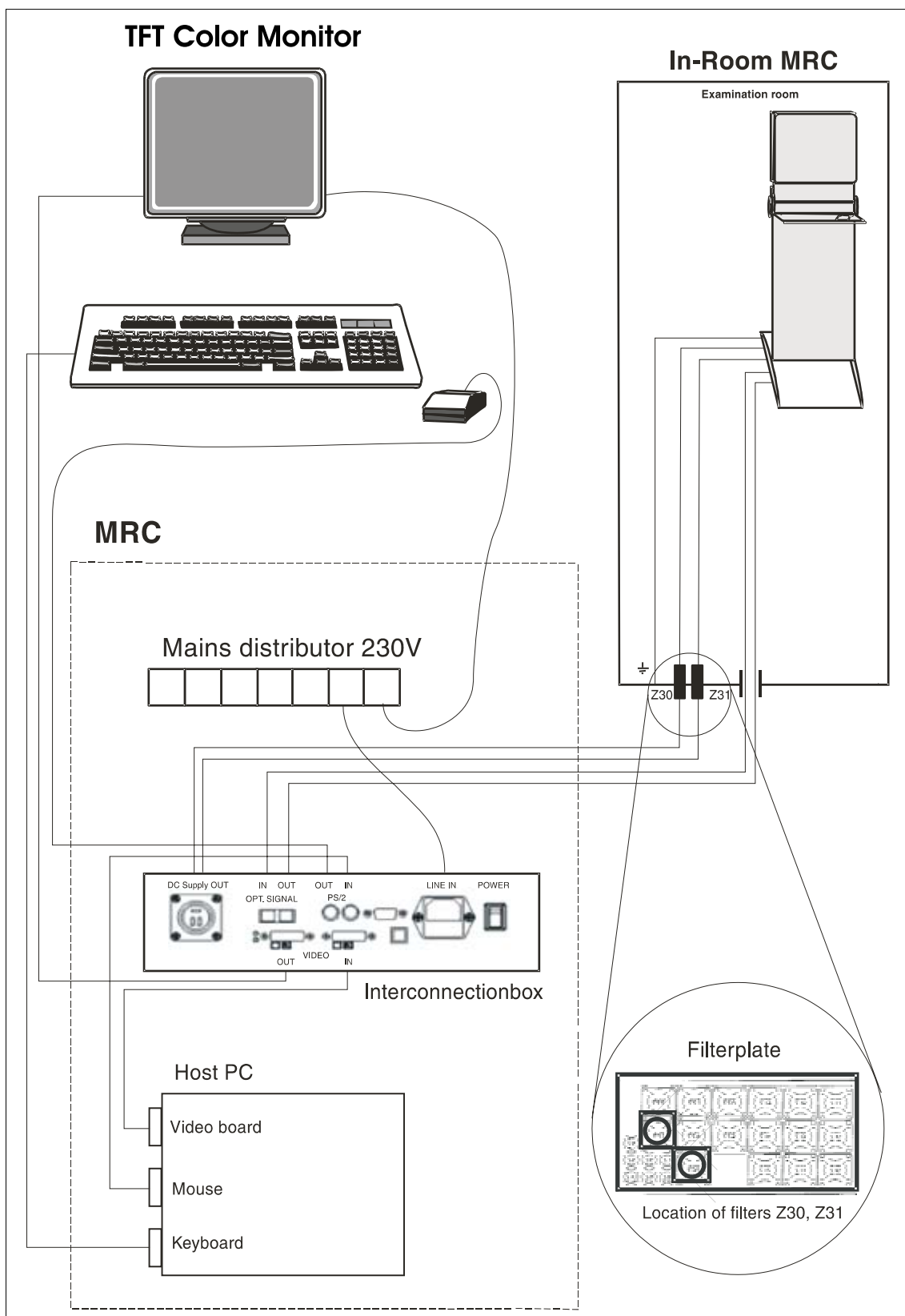


Fig. 51: Cabling with option In-Room MRC

Start-up procedure for additional monitors (optional)

Presettings and voltage adjustment 230V/115V for video multiplexer and signal converter.



Fig. 52: RGB VideoLink

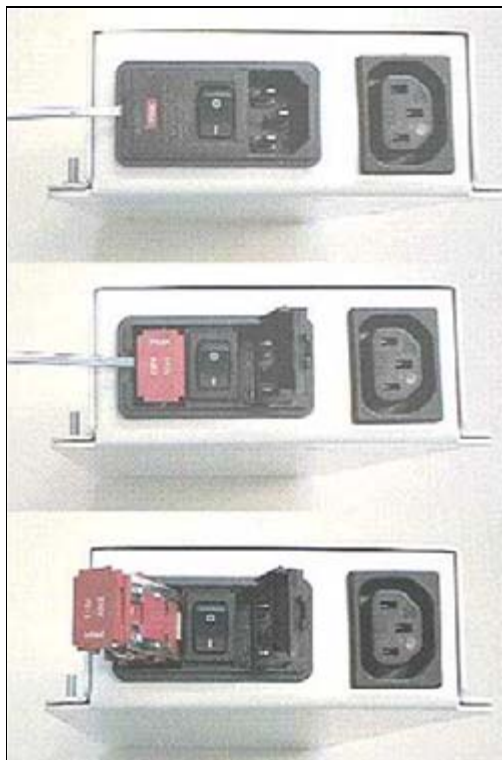


Fig. 53: Voltage adjustment

- Verify the correct voltage adjustment prior to switching on the monitor.
- If you need to change the voltage from 230V to 115V, refer to the following steps:
- **Switch off the main power and disconnect all power and signal cables.**
- The mains inlet with the voltage switch is located on the left side of the video multiplexer.
- Carefully open the cover with a screwdriver.
- The voltage switch is now visible.
- Carefully lift the voltage switch with the screwdriver and remove it completely.
- Turn the voltage switch 180° around its longitudinal axis so that you can see the required voltage on the left side.
- Insert the voltage switch. Close the cover. The required voltage value should now be visible through the small frame.

- Set both the “**Sync on Video**” and “**Compensation Output**” switches of the video multiplexer to **OFF**. Switch on the main switch.



Fig. 54: Video multiplexer

- Pos. 1 Main switch
Pos. 2 Sync on video/compensation output

NOTE

Do not adjust the voltage at the video multiplexer.

Clone Mode

Setting the Clone Mode

NOTE

The Clone Mode is necessary, if an additional (second) monitor is connected to the second video output of the graphic card (for both digital or analog signals).

In the Clone Mode, the original Windows desktop from the primary VGA display (primary monitor) is replicated onto the secondary VGA display (secondary monitor, e.g. Remote Monitor).

The following procedure has to be performed for the additional monitor when:

- Only one graphic card is in use.
- The clone mode is not activated.
- The additional monitor receives the video signal from the second video connector of the graphic card.

Preparation

Connect to the second graphic card output connector (2/Fig. 55 / p. 75) to receive the required (analog or digital) video signal.

The following picture shows the graphic card connectors:



Fig. 55: Graphic card connector

Pos. 1 DVI connector for the digital TFT monitor (monitor 1)

Pos. 2 Second video connector (e.g. for the remote monitor) can provide an analog and digital video signal, here with standard VGA adapter for an analog signal

- Switch on the monitor, turn on the PC.
- Login as administrator.
- Right-click the the background of the desktop and select properties
or press buttons Ctrl and Esc simultaneously - the Start button appears
Click: Settings -> Control Panel select and click -> Display
Click the **Settings** tab card in the pop-up window
Click button **Advanced** and choose the **Graphic-Card**tab card (in the example
([Fig. 56 / p. 76](#)) it isQuadro FX 3400)

- Select the Display Settings, in the presented example it is: nView Display Settings (1/ Fig. 56 / p. 76)

Then navigate to clone mode selection (in the example navigate to the select-box of nView Display Mode) (2/ Fig. 56 / p. 76)

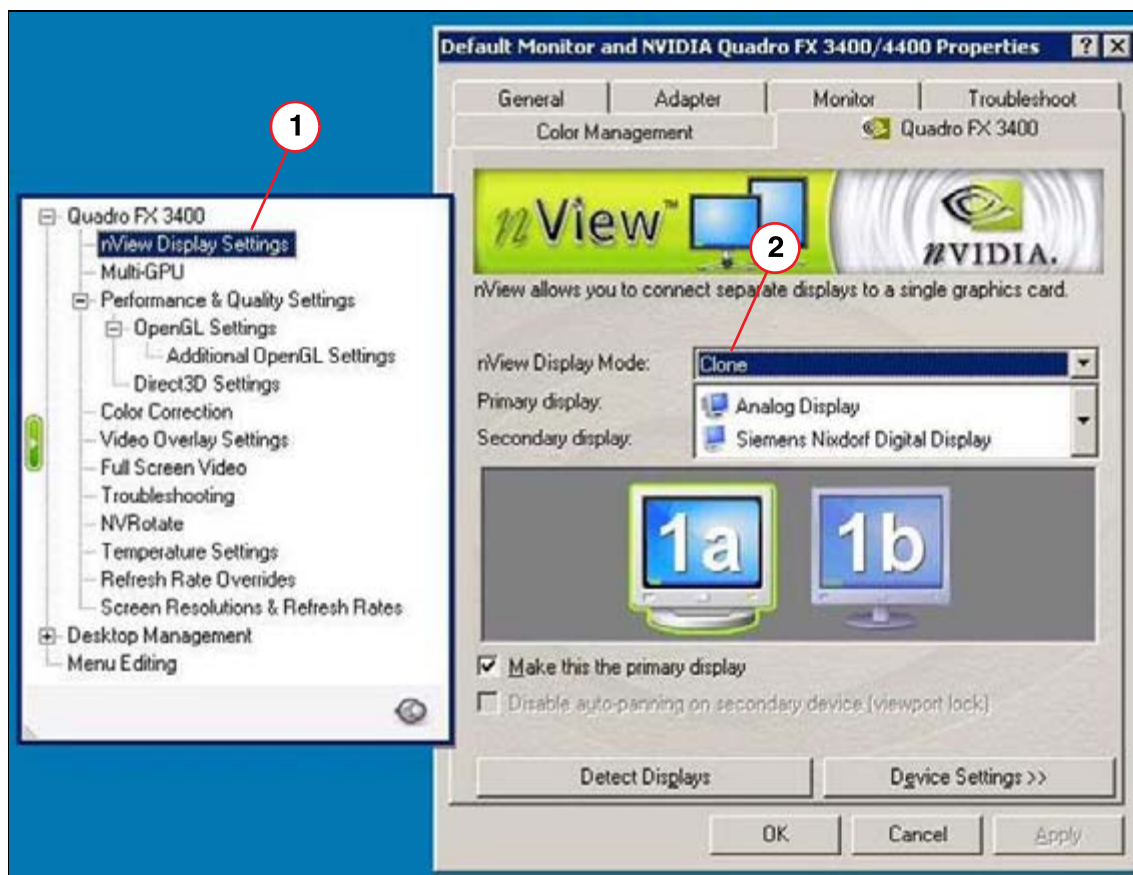


Fig. 56: Graphics card settings (example)

Pos. 1 nView Display Settings

Pos. 2 nView Display Mode

Accept the new settings and the second monitor should show exactly the same information on the display as the first monitor.

NOTE

The monitor adjustment has to be checked and, if necessary, it has to be readjusted.

Appendix

Visually inspecting/evaluating the TFT monitor

Prior to evaluation, the TFT display must be in operation for at least 20 minutes (warm-up).

Checking/preparing the settings of the graphics card

- Enable the Advanced User functionality:
 - Simultaneously press buttons **Ctrl** and **ESC** - the Start button appears
 - Click: **Start** -> **Programs** -> **Advanced User** -> enter **PW** -> press **OK** button
- The graphics card operates in **VESA SXGA / 1280 x 1024** mode at **75 Hz**.

NOTE

TFT's display flicker-free pictures as well with a refresh rate of 60 Hz.

- Display the Windows XP screen as follows:
 - Simultaneously press **Ctrl + Alt + Del**
 - Select the **Task Manager**
 - **Right-click** the **CapGM-Exam** in the list
 - In the new pop-up window click **Minimize** (WIN XP Desktop is now displayed in foreground)
- Check the **VESA mode** and **screen refresh rate**:
 - Right-click the background -> select **Properties**
 - Click the **Settings** tab card in the pop-up window
 - Check **VESA mode 1280 x 1024** ([Fig. 57 / p. 78](#))
 - Click the **advanced** button and choose the **Monitor** tab card
 - Check the **screen refresh rate of 75 Hz** ([Fig. 58 / p. 78](#)) (60 Hz for TFT's)
 - Correct the setting, if necessary
 - Confirm each pop-up window by clicking ok

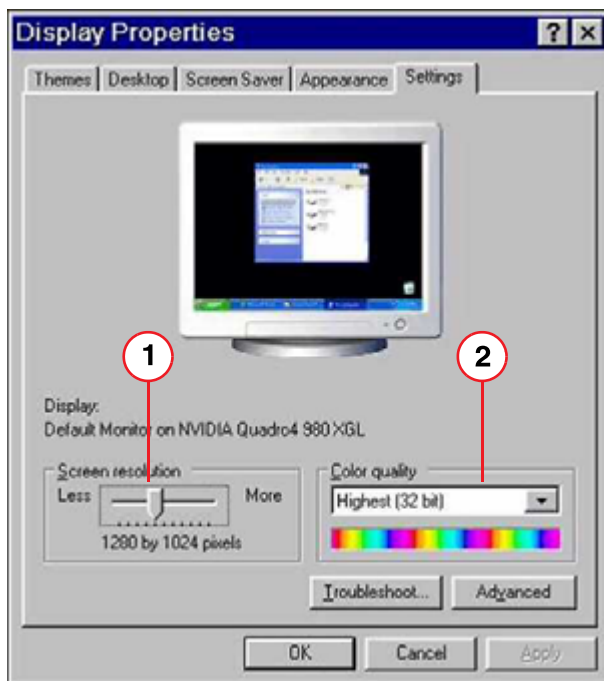


Fig. 57: VESA mode

Pos. 1 1280 x 1024

Pos. 2 True Color

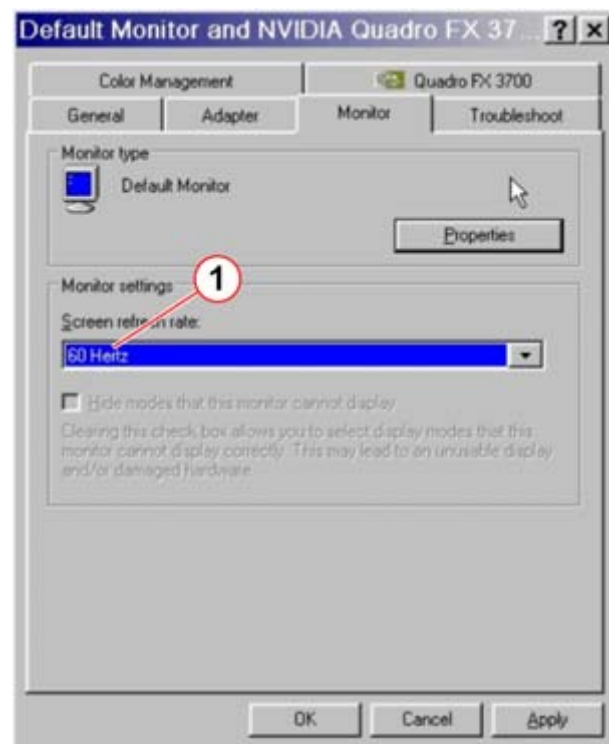


Fig. 58: Screen refresh rate

Pos. 1 60 Hz.

NOTE

The type of pop-up window depends on the type of graphics card used. It can differ from the examples (Fig. 57 / p. 78) and (Fig. 58 / p. 78)

Preparation for visual evaluation**NOTE**

The display must be in operation for at least 20 minutes.

NOTE

The ambient light and monitor position must correspond to the operating conditions (to evaluate the criteria for contrast and patient images on-screen, the ambient light has to be at a dark level) and must be reproducible (this should be specified together with the customer).

- If the NUMARIS application is displayed again, minimize it as described above (task manager); result: the test image is displayed in full-screen mode (Fig. 59 / p. 79).
- Call up the **test image Master2ALLRev20.bmp** as follows:
- **Right-click** the desktop background.
- Click **Properties** and choose the **Desktop** tab card.
- Click **Browse**.

- Select the path:
C:\MedCom\utils\Imagequality\Master2AllRev20.bmp
- Click **Open** and **OK**, the result shows (Fig. 59 / p. 79).

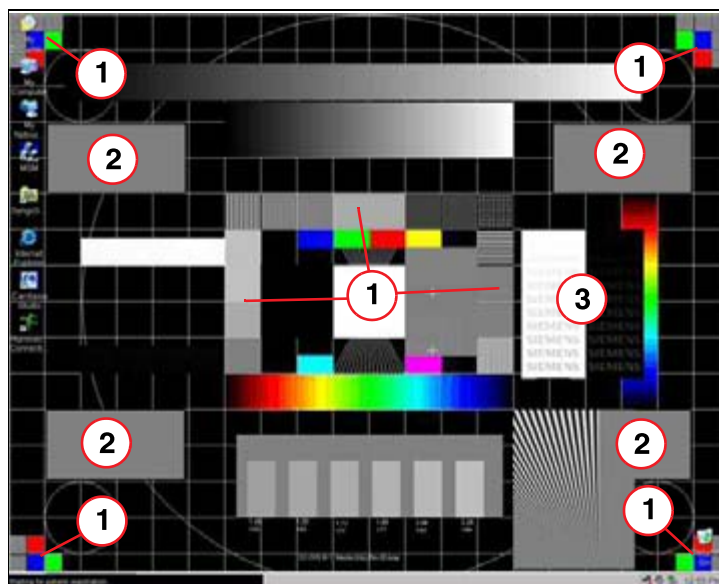


Fig. 59: Test image Master 2 All Rev 20

- Pos. 1 High frequency line pattern groups
 Pos. 2 Pixel on/off fields
 Pos. 3 SIEMENS logos (black and white)

NOTE

Before starting to evaluate the monitors with regard to the image quality criteria, the Auto adjust program has to be run as described for monitor type B, monitor C, and TFT monitor DSC 1904-D, DSC 1907-D. Refer to the section "Adjustment procedure" in Replacement of Parts.

Visually checking the geometry

Evaluation criteria for the geometry

- The image should be centered and fill the display area to the edge; i.e. the image should be equidistant to all edges of the usable screen area.
- All screen information has to be visible along the edges of the displayed image; i.e., the double white frame must be visible on all sides of the screen.
- The horizontal/vertical ratio should be correct (squares must appear as squares, circles as circles).
- Grid lines and other fields must not be distorted.
- If one or more of these criteria are not met, proceed with **monitor type B**, **monitor type C** and **TFT monitor DSC 1904-D, DSC 1907-D** as described in the **Adjustment procedure** section in Replacement of Parts.

Visually checking the frequency and the phase

- The **Pixel on/off** fields (Fig. 59 / p. 79) must appear as uniform and smooth areas.
- No flickering or horizontal interference lines should be seen (Fig. 60 / p. 80) and (Fig. 61 / p. 80).
- The high-frequency line pattern groups of the test image (Fig. 59 / p. 79) should be visible in both the horizontal and vertical directions and should not have colored edges.

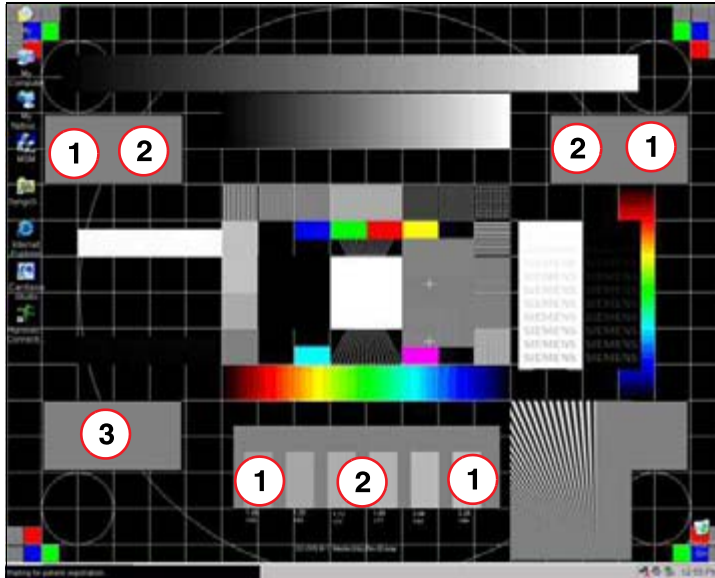


Fig. 60: Example 1: Misadjustment of frequency

- Pos. 1 Dark region
 Pos. 2 Bright region
 Pos. 3 Zoomed region (horizontal jitter)



Fig. 61: Zoomed region (horizontal jitter)

- Call up a text file and maximize it, such as C:\MedCom\log\Startup.log.

- No vertical interferences should occur (Fig. 62 / p. 81).

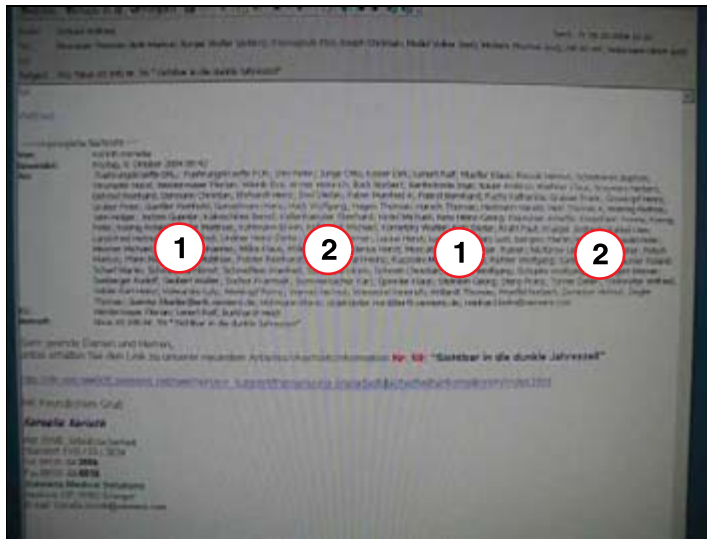


Fig. 62: Example 2: Misadjustment of frequency

- Pos. 1 Characters are not sharp
Pos. 2 Characters are sharp

- All characters are sharp.
- All edge and grid lines must be clear and sharp.
- If one or more of these **criteria are not met**, proceed as per the directions in the Adjustment procedure section for **monitor type B, monitor C, and TFT monitor DSC 1904-D, DSC 1907-D** in Replacement of Parts.
- Close the text file.

Visually checking the contrast

Evaluation criteria for contrast setting

IMPORTANT:

- All 10 'SIEMENS' logos in the black and white fields (Fig. 59 / p. 79) must be visible. The ambient light should correspond to the actual operating conditions on-site (this should be reproduced in consultation with the customer).
- The gradations in the upper gray scale should not be seen as steps.
- If one or more of these **criteria are not met**, proceed as per the directions in the Adjustment procedure section for **monitor type B, monitor C, and TFT monitor DSC 1904-D, DSC 1907-D** in Replacement of Parts.

Checking the luminance

Preliminary step:

- The Siemens test image Master2ALLRev20.bmp (Fig. 63 / p. 82) is called up as the desktop background.

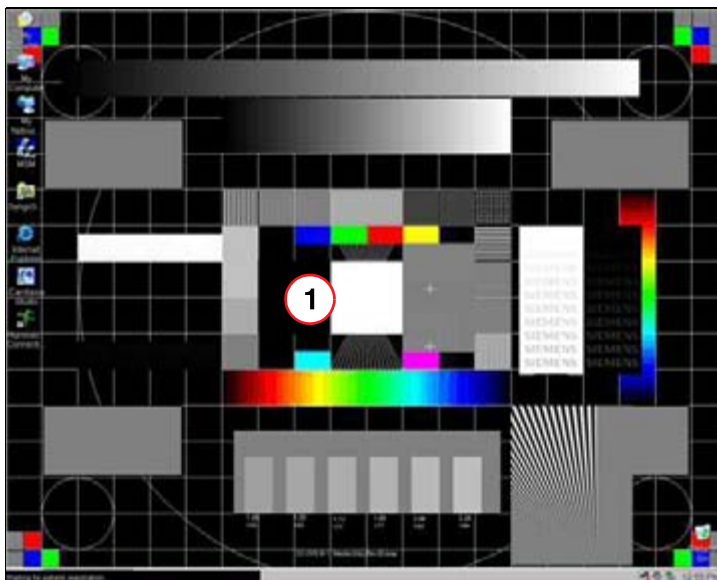


Fig. 63: Test image Master 2All Rev20

Pos. 1 Measurement fields for 0% and 100% values

- The **white and black test fields** in the center of the test image are used to **measure** and **adjust** the image (marked in (Fig. 63 / p. 82)).

Preparing the SMFitAct



Fig. 64: SMFit Act with diffusion screen

Pos. Membrane switches

NOTE

Each time the SMFitAct is switched on, it must be set to zero ([Zero adjustment / p. 83](#)).

Zero adjustment

The tube extension must be removed (unscrewed) for **direct measurement**.

- Press the **on/off** button, the display reading should be stable and not flashing.
- The letter '**D**' should be shown in the upper left corner of the display (meaning: **direct** measurement). If an '**A**' is displayed instead, press the **on/off** button again.
- Cover the SMFitAct tube completely, for example set it on a level surface (table) with the diffusion screen attached ([Fig. 64 / p. 83](#))
- Press both the **on/off** and **run/stop** switches at the same time.
- The value **0.00** should be displayed.

Additional instructions for operation can be found in the accompanying manual.

Measurement procedure

NOTE

When performing contact measurements using the spotmeter SMFit Act (with diffusion screen attached), ensure that you do not exert any pressure on the display panel while using the sensor. Otherwise, panel damage or malfunction may result.

Measurement of the white field (100% value)

- Without applying any pressure, center the diffusion screen in the 100% field ([Fig. 65 / p. 84](#)). The **center hole** should be **filled completely** by the **100%** field. Carefully slide the SMFitAct probe through the center hole of the diffusion screen ([Fig. 66 / p. 85](#))
- Hold the spotmeter **exactly at right angles** to the screen, otherwise the measurement **values are not reliable**.

Read the 100% value and enter the data in ([Tab. 11 / p. 87](#)) (100% - measured value - pre-adj.). The following measurement tolerances are permitted:

Recheck: $137 \text{ cd/m}^2 (+8 / -7) \text{ cd/m}^2$

Adjustment $137 \text{ cd/m}^2 (+3 / -2) \text{ cd/m}^2$

NOTE

If setting values have been changed using the OSD menu, wait until the reading of the TFT display stops fluctuating (typically 5 to 10 seconds). The adjustment requires a certain amount of time to reach the desired state.

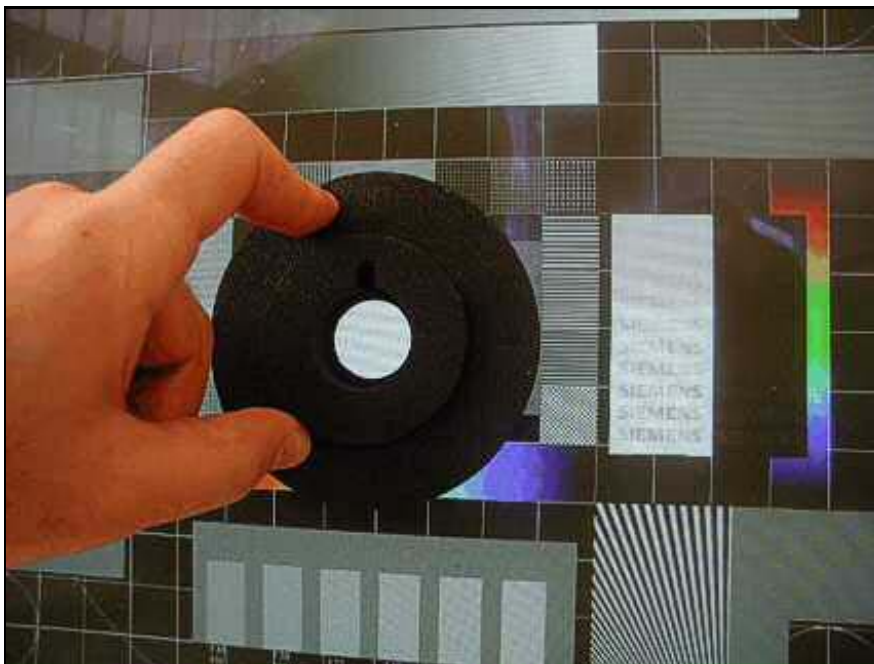


Fig. 65: Diffusion screen positioning at 100%

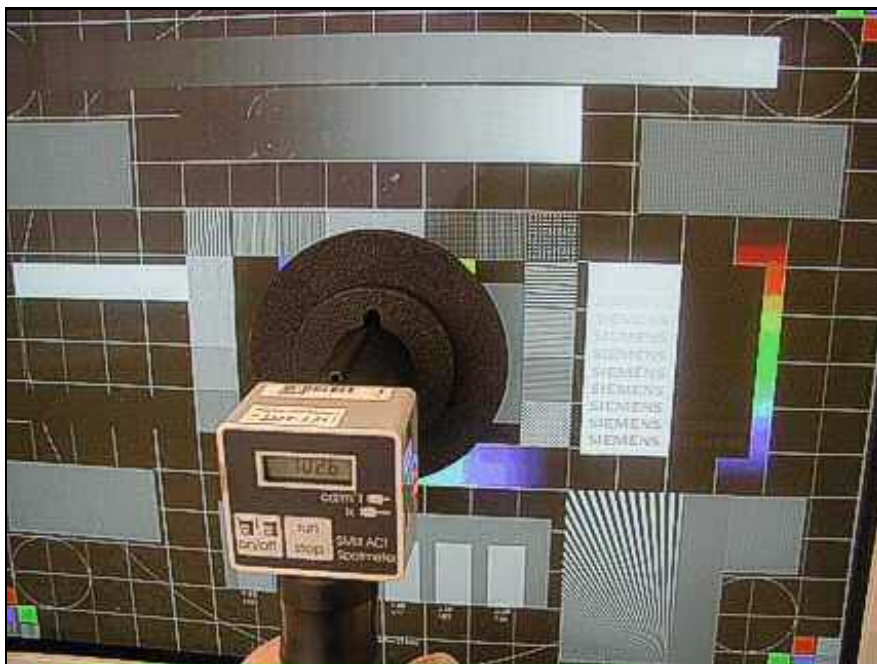


Fig. 66: Measurement position for 100%

Measurement of the black field (0% value)

- Proceed as with the 100% field, but use a different measurement position (Fig. 67 / p. 86) and (Fig. 68 / p. 86). The following measurement tolerances are permitted:

Recheck: $0.5 \text{ cd/m}^2 (+/-0.2) \text{ cd/m}^2$

Adjustment $0.5 \text{ cd/m}^2 (+/- 0.1) \text{ cd/m}^2$

- Enter the measured value in (Tab. 11 / p. 87) under: 0% - measured value - preadj.

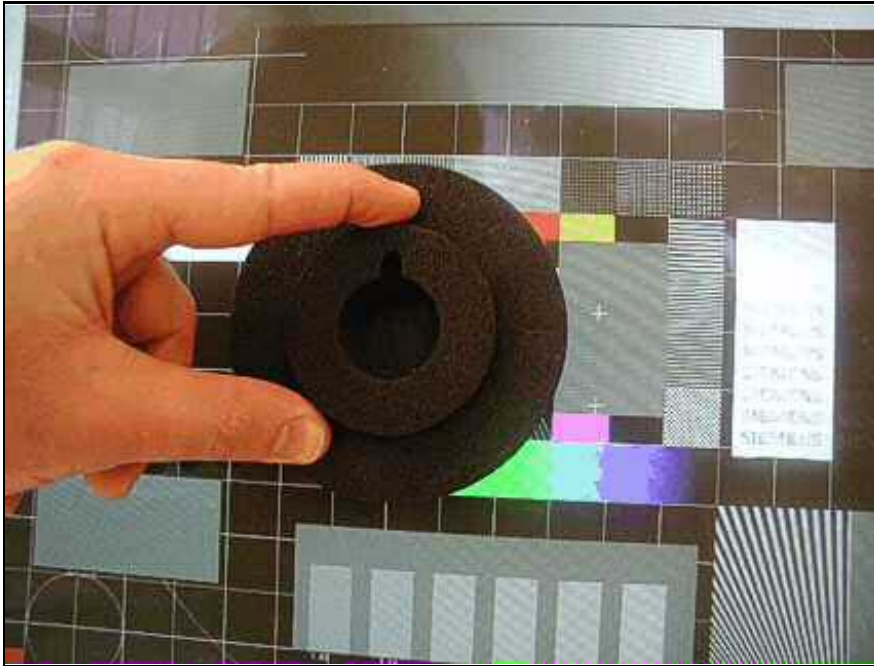


Fig. 67: Diffusion screen positioning at 0%



Fig. 68: Measurement position for 0%

If these tolerances (adjustment of the values of 0% and 100%) and the test criteria described above have been met, the monitor has already been adjusted to the corresponding PC. There is therefore no need for adjustment.

Tab. 11 Contrast, Brightness, Backlight

Monitor type: Serial no.:	MRC				MRSC			
	Measured value		Slider position		Measured value		Slider position	
	Preadj.	Correct.	Preadj.	Correct.	Preadj.	Correct.	Preadj.	Correct.
100% /Contrast								
0% /Brightness								
Backlight:	-----	-----			-----	-----		
Frequency	-----	-----			-----	-----		
Phase/Phase shift	-----	-----			-----	-----		

- If one or both measured values are not within the tolerance level, proceed with **monitor type B**, **monitor type C**, and **TFT monitor DSC 1904-D**, **DSC 1907-D** as described in the **Adjustment procedure** section in Replacement of Parts.
- In the case of problems, refer to the **Appendix in Replacement of Parts**.

NOTE**Reset the background of the screen to normal**

- Click the right mouse button once, a window pops up.
- Click the menu item **Properties**.
- Choose the **Desktop** tab card.
- Select **None** from the list and confirm with **OK**.

1. Remedy for misadjustment of brightness / contrast

Total misadjustment of brightness and / or contrast, all 10 black and 10 white 'SIEMENS' logos and/or the contrast fields (5% and 95%) of the SMPTE test pattern are not fully visible.

- Open the **OSD** menu.
- Set the **brightness** to **50%**.
- Set the **Contrast** to **Maximum**.
- Measure the **100%** field using the **SMFitAct**.
- Record the measurement value for the **Backlight** parameter (**preadjusted**) in (Tab. 11 / p. 87).
- Adjust the **Backlight** to $137 \text{ cd/m}^2 +3/-2 \text{ cd/m}^2$.
- Record the **new slider position** of the parameter **Backlight (corrected)** in (Tab. 11 / p. 87).
- Decrease the **Contrast** until its value starts to vary (decrease).
- Then slightly increase the **Contrast** until the adjusted value for **Backlight** is met again.

- Now **all 10 'SIEMENS' logos** within the **white field** or the **95% field** of the **SMPTE** test pattern should be completely **visible**.
- If not, adjust the **Backlight (100% field)** within the specification limits **135 cd/m² - 140 cd/m²** and readjust the **Contrast** until the evaluation criteria and specifications are met.
- Measure the **0% field**.
- Decrease the **Brightness** until it stops varying (typically at **0.3 - 0.4 cd/m²**).
- Slightly increase the **Brightness** until the specifications for the **0% field** of **0.5 cd/m² - 0.4 cd/m² +/- 0.1 cd/m²** are met.
- Now **all 10 'SIEMENS' logos** within the **black field** or the **5% field** of the **SMPTE** test pattern should be completely **visible**.
- If not, adjust the **Brightness (0% field)** within the specification limits **0.4 cd/m² - 0.6 cd/m²** until the evaluation criteria and specifications are met.
- Check the measurement values for the **100% field** again.
- If the specifications are met, **record the corrected measurement values and slider positions** of the parameters **Contrast** and **Brightness** in (Tab. 11 / p. 87).
- If the **specifications cannot be met**, reset the factory defaults as described below (4. **General: adjustments failed**) and repeat to perform a complete monitor adjustment.
- If specifications are still not met, replace the monitor.

2. Resetting the factory defaults

Before replacing the TFT monitor due to misadjustment, you are able to reset the factory defaults for both monitor types so that you can restart the adjustments under defined conditions:

- For monitor **type 'B'**, follow the instructions below:
 - Call up the **OSD** menu
 - Select **Others -> User Setting -> Service**
 - Press the Up key once and the Down key twice - a new window pops up
 - Select **User -> Reset User Setting**
 - Perform a complete adjustment of the TFT monitor
- For monitor **type 'C'** perform the following steps:
 - Prerequisites: Call up the **OSD** menu. The factory parameter settings should have been entered already in the dedicated table of the **Start-up** and/or in the **Replacement of Parts** documentation.
 - Resetting the factory defaults: **OSD main menu -> Reset >>>**.
 - Check the parameter **Backlight** and manually readjust it to the original value, if required.
 - Perform a complete adjustment of the TFT monitor.
- Not applicable for TFT monitor type DSC 1904-D
- For TFT monitor type DSC 1907-D, refer to the OSD menu description.

3. Missing color, color overtone or incorrect setting for screen color

- Check the position of switches for the video input, ensure that the termination switches for **RGB** are in the **75 Ohm** position.
- Check the **Color/black-white** setting for **type 'B'**:
 - Set the **Color** mode at the RGB/sw switch located in back of the monitor to position **RGB**.
 - Call up the **OSD** menu, open **Brightness Contrast** and check the setting of the **Color** parameter.
 - By default **button 2** is activated and is **illuminated green**.
- Check the **Color/black-white** setting for monitor **type 'C'**:
 - Open the **OSD** menu.
 - To set the **Color** mode, uncheck the **Monochrome** menu item.
- Check the **Color/black-white** setting for monitor **type 'DSC 1904-D and DSC 1907-D'** using the OSD menu.

4. Automatic adjustment of geometry failed (for monitor type 'B' only)

If automatic adjustment of geometry failed (for monitor type 'B' only) and the satisfactory of sharpness is not obtained:

- Check the slider position for the **Frequency** parameter.
- Open the **OSD** menu, select menu item **Others**, activate submenu **Frequency Phase**.
- Toggle to parameter **Frequency** and set the slider to the typical value range from **60** to **80** (a typical value is: **64**) and check all criteria indicated ([Visually checking the geometry / p. 79](#)).

BIOS CD update and installation

General

This chapter describes how to download and install the current BIOS CD.

NOTE

A new release of the BIOS CD is published via speedinfo.

Prerequisites

For downloaded files

Material no.	Name
03 07 86 31	Blank "write once read many" (WORM) CD media TDK CDR 74 or equivalent.
n/a	SPC equipped with CD-R drive (CD burner)
n/a	Up-to-date virus scanner on SPC
n/a	Burning software such as Nero or equivalent on SPC
06 00 14 86	Blank floppy disk 3.5" 1.44MB - Sony 10MDF-2HDCF or equivalent

Intranet download preparation

Get the new version of the BIOS CD (name is given in the speedinfo). Download the self-extracting file "BIOS_CD_xxxxxx.exe" from the MED CS Intranet.

- Navigate to
["https://intranet.healthcare.siemens.com/cms/CS/en/Home/Pages/Default.aspx"](https://intranet.healthcare.siemens.com/cms/CS/en/Home/Pages/Default.aspx)
 and follow the path **For Service > MR Systems > Downloads > "SoftwareVersion"**.
 The **SoftwareVersion** is given via the speedinfo.
- Download the self-extracting file "BIOS_CD_xxxxx.exe" to your local PC, save the file to the hard drive c:\... and close the window when finished.
- Check the file with a standard virus scanner.
- Double-click the downloaded file. Extract the file by clicking "Unzip". Check if the files have been successfully extracted (pop-up message) and click Close to continue.
- Label a blank CD-R as follows, "MR005 BIOS-CD VBxx : xxxxxx" and insert it into the CD burner drive of the SPC (Service Personal Computer).
- Open Windows Explorer.

- Navigate to the directory where the files were extracted, in this case
C:\temp\BIOS_CD_xxxxxxRun the batch program cdrec_BIOS_CD.bat, and double-click it. The contents of the ISO file including the boot information will now be automatically written to the CD.
- Wait until the CD is ejected from the drive.
- Press any key (e.g. space) to continue.
- Remove the CD from the drive and close the drive.

NOTE

Once the files are extracted to the CD/floppy disk, these files cannot be copied to another CD/floppy disk for distribution (e.g. no CD to CD copies as it is impossible to verify whether the SW is free of viruses).

BIOS update on MRC

ACTION	MRC
• During the boot-up process, press 'F2' repeatedly until the BIOS set-up appears. For the BIOS password, refer to speedinfo MR_25/04 in the CS Knowledge Base.	☐
• Only for R630: Go to Security and check if the entry "Start from IDE CD-ROM" is set to Enabled.	☐
• Go to Boot -> Boot Device Priority and move CD-ROM Drives to the top by selecting the "CD-ROM Drives" entry and pressing the + key.	☐
• Insert the CD titled "MR005 BIOS-CD VBxx : xxxxxx" into the CD/DVD-ROM drive (R:).	☐
• Press F10 and Enter to save and exit the BIOS.	☐
• The BIOS CD is booted and the BIOS updates run automatically; no user action is necessary.	☐
• The BIOS update ends when the following information is displayed BIOS Installation Done - System Ready for Software Installation Please Remove the BIOS CD ! Please Turn Off Power or the Reset System ! C:\>	☐
• Remove the "MR005 BIOS-CD VBxx : xxxxxx".	☐
• Press the key combination <CTRL> + <ALT> + (on a German keyboard use <Strg> + <Alt> + <Entf>) to reboot.	☐

BIOS update of the MRIR

Perform the BIOS update of the MRIR!

Steps at the MRC

NOTE

Be sure that the network cables on the MRC and MRIR are connected. If not, the BIOS update will fail, because the BIOS update is now performed via the network.

ACTION	MRC
To run the BIOS update for the MRIR, you have to be logged in as Administrator. If necessary, reboot and log in as Administrator (disable the auto login function in the local service UI). Note: The BIOS update for the MRIR can be done only under "user administrator."	☐
Insert "MR005 BIOS-CD VBxx : xxxxxx" into the HOST CD/DVD-ROM drive (R:\).	☐
Start Windows NT Explorer: Click START -> Programs -> Accessories -> Windows Explorer.	☐
Navigate to the CD-ROM drive R:\BootNet\Host.	☐
Run the BootNet.bat program by double-clicking it.	☐
Restart the imager as requested in the pop-up message. Switch the imager off and on. Note: The BIOS update for the MRIR can be done only under "user administrator." The OS for the BIOS update is booted from the imager MRIR via the network. The OS and the BIOS update firmware and settings are on the BIOS CD in the CD/DVD-ROM drive of the host MRC. Do not remove the BIOS CD from the CD/DVD-ROM drive. If the BIOS update does not finish and the BootNet.bat program is terminated by the user (CSE), the following program must be executed: R:\BootNet\Host\BootNetRel.bat (see also workaround next page)	☐
Wait until the BIOS update process is completed (duration is up to 5 min.).	☐
A pop-up with an 'Okay' or 'Yes' button appears; press okay before you continue.	☐
Remove the "MR005 BIOS-CD VBxx : xxxxxx" from the CD-ROM drive.	☐
Shut down and switch off the MR system at the alarm box.	☐
Switch on the MR system at the alarm box.	☐
Wait until the MR system is up and ready for measurement.	☐

Hints and workarounds:

BIOS CD logfile

The logfile for the BIOS update of the imager is

`C:\Temp\BIOS_CD\BiosCD.log`

The logfile for the BIOS update of the high-end imager is

`C:\Temp\BIOS_CD\TY*****.log`

It is possible for the BIOS update on the imager to fail. In this case, a log file with an error message is automatically displayed at the MRC. If this happens, proceed as follows. If the BIOS update fails or the BootNet.bat program was terminated by the user (CSE) the following workaround must be done:

1. Run the following program from the BIOS CD:

`R:\BootNet\Host\BootNetRel.bat`

2. Perform the BIOS update again.

Procedure for reinitializing the MRIR

- Problem description:

The MRIR is not coming online or is not functional (MRIR symbol is crossed out). The OS is booted and the network communication (ping) is OK. With the following steps you can reinitialize the software configuration on the MRIR.

- Solution:

The following file has to be deleted:

`C:\MedCom\M CIR\Med\Config\scripts\setup.lastcall`

In order to delete this file following the procedure:

- Enable advanced user.
- Start Windows Explorer.
- Delete the file `C:\MedCom\M CIR\Med\Config\scripts\setup.lastcall`.
- Shut down the MR system and switch it off at the alarm box.
- Switch on the MR system at the alarm box.
- Wait until the MR system is ready for measurement.

Final checks

NOTE

Check if hotfixes and service packs are available for this software version according to the various speedinfos and update instructions.

- Perform the “Quality Assurance” procedure: “Image Orientation” in the service platform.
- Make sure that the system is generating images.

⇒ NEW: Replacement of fans, power supply, and battery.

Revision 03

- Chapter 03
 - Editorial changes